

Better Jobs for Mothers, More Sons at Birth*

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Abstract

Vietnam's sex ratio at birth became more male-biased even as gender gaps in employment and education narrowed. We test a household model with son preference in which maternal wage gains tighten the time cost of childbearing, lowering fertility and raising sex selection despite higher returns to daughters. Exploiting industry-level tariff cuts from the 2001 US-Vietnam trade agreement, we assign exposure separately to each parent. Maternal exposure raises male births and lowers fertility, whereas paternal exposure raises fertility but not sex selection. Closing labor-market gender gaps can widen gaps at birth where childcare falls mainly on mothers and son preference persists.

JEL Classification: F13, J13, J16, J22

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1. Introduction

Vietnam’s sex ratio at birth rose from 107 boys per 100 girls in 1999 to 112 by 2015. Over the same period, female wage employment nearly doubled as a share of total female employment, fertility remained low, and women’s university enrollment rose relative to men’s (Figure 1). Similar combinations of rising sex ratios and expanding labor-market opportunities for women appear in other countries with son preference. An influential view holds that women’s economic gains often improve child outcomes and can benefit daughters in particular (Jayachandran and Voena, 2026; Duflo, 2003; Qian, 2008). Why, then, can closing gender gaps in jobs and schooling coincide with a widening gender gap at birth?

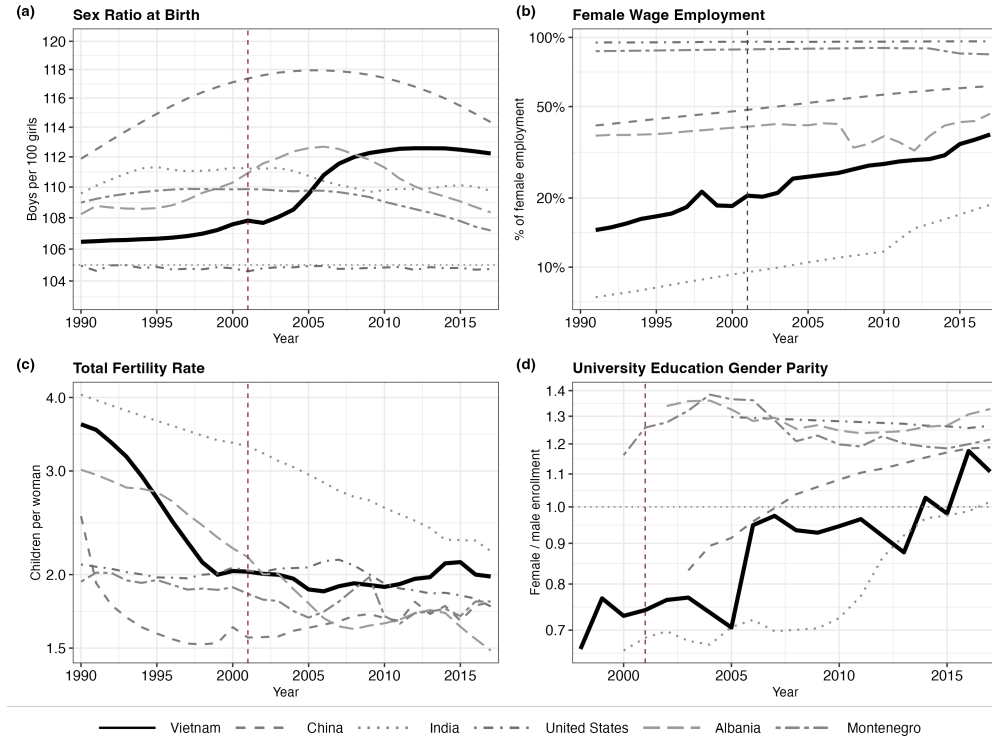
The sex-selection literature points in several directions, and the evidence is mixed. Fertility decline can raise the sex ratio when families with son preference have fewer births over which to realize a desired son (Jayachandran, 2017). Fertility restrictions can intensify the same pressure, as evidence from China links stricter fertility limits under the one-child policy to more male-biased sex ratios (Li et al., 2011; Ebenstein, 2011). A different prediction arises when economic opportunities improve for women. If these gains raise the economic value of daughters, relative-returns arguments predict that sex selection should fall (Rosenzweig and Schultz, 1982; Qian, 2008). By contrast, income gains can worsen sex ratios when sex selection rises with household resources, as in China’s land reform (Almond et al., 2019).¹ These mechanisms imply different responses to rising female employment, but the evidence often comes from shocks assigned at the location or household-income level. Such variation cannot distinguish whether a response comes from mothers’ labor-market opportunities, fathers’ income, or broader local gains. Without that separation, fertility decline, relative returns, and income remain difficult to distinguish empirically.

This paper separates these channels by comparing labor-market gains that reach mothers and fathers differently. We show that maternal labor-market gains increase the male share at birth, while paternal labor-market gains do not. We explain the contrast through a household model in which mothers bear childcare time costs and parents have son preference. The fertility literature has long emphasized that higher female wages can reduce fertility because childcare remains disproportionately maternal (Doepke et al., 2023). In a setting with sex selection, this familiar mechanism has an additional implication. When maternal wages rise, children become more expensive in time, fertility falls, and parents with son preference have stronger incentives to concentrate remaining births on sons. When paternal wages rise, household resources increase without tightening the maternal time constraint. The model therefore provides a microfoundation for the fertility-decline mechanism emphasized by Jayachandran (2017), while distinguishing that channel from income and relative-returns explanations.

Our model also explains why rising investment in girls need not imply less sex selection. We

¹Marriage-market channels provide another mechanism in settings where relative wealth and marriage payments are central (Borker et al., 2026). We do not model this channel because it is not first-order in Vietnam, where dowry is uncommon.

Figure 1: Sex Ratio, Female Wage Employment, Fertility, and Education Gender Parity



Notes: Data from [Chao et al. \(2019\)](#) (panel a) and the World Bank World Development Indicators (panels b–d). Vietnam is the solid black line. Vertical dashed line marks 2001 (BTA). Horizontal dotted line in panel (a) marks the natural sex ratio (≈ 105). Panel (d) is the Gender Parity Index for university enrollment (female-to-male ratio). Log scale on all vertical axes.

combine the time-cost margin with the quantity-quality tradeoff ([Becker and Lewis, 1973](#)). Higher returns to daughters could in principle reduce sex selection by raising the value of daughters ([Qian, 2008](#)). But these higher returns also raise desired investment per daughter. Panel (d) of Figure 1 shows the relevant fact for Vietnam: the female-to-male university enrollment ratio moves toward parity over the same period in which the sex ratio at birth becomes more male-biased. In the model, the narrowing education gap makes children more expensive to rear, reinforcing the fertility decline already operating through the time-cost margin. Under son preference, the extra fertility decline produces more male-skewed births. Combining the two margins shows how maternal labor-market gains, narrowing education gaps, and rising male shares at birth can coexist in the same setting.

We test these predictions using the 2001 US-Vietnam Bilateral Trade Agreement (BTA), which reduced US tariffs on Vietnamese exports by different amounts across industries. These industry-specific export-demand shocks reached mothers and fathers differently depending on where each parent worked. We combine the 1999 Vietnam Population Census with the Vietnam Household

Living Standards Surveys (VHLSS, 2002–2010), which record each parent’s industry and employment close to the time of birth. We restrict attention to children aged 0 and 1, so the observed sex ratio closely tracks the sex ratio at birth and is less affected by differential mortality, migration, or later household composition changes. Observing both parents’ industries at the time of birth is the key advantage of these data. Prior studies of sex selection often rely on censuses or retrospective birth histories that recover longer fertility series but cannot assign contemporaneous industry to both parents for each birth. If parents switch industries or migrate after conception, later measures make it harder to distinguish maternal exposure, paternal exposure, and broader provincial exposure. The VHLSS gives a shorter but sharper window. It identifies parent-specific exposure close to birth, allowing us to separate shocks operating through the mother’s job, the father’s job, and the broader local economy.

We find that sex selection responds to the mother’s labor-market exposure but not to the father’s or the province’s. At the median industry tariff cut, roughly a 7 percentage-point reduction, maternal exposure raises the probability of a male birth by about 2 percentage points, a gap of roughly 9 boys per 100 girls between exposed and unexposed mothers. A 10 percentage-point larger tariff cut in the mother’s industry also raises her probability of holding a wage job by 6.4 percentage points, about a quarter of the 28 percent sample mean. In annual fertility histories reconstructed from household rosters, maternal exposure lowers the probability of a birth while paternal and provincial exposure raise it, and neither fathers’ industries nor the local labor market moves sex selection. The pattern across mother, father, and province matches the model’s central prediction that only income arriving through the mother’s own job tightens the maternal time constraint.

The paper contributes to three literatures. First, the model reconciles findings in the sex-selection literature that appear to point in opposite directions. In China, [Almond et al. \(2019\)](#) show that income gains increased sex selection, while [Qian \(2008\)](#) finds that higher female-specific returns reduced it. Evidence on the one-child policy also shows how fertility constraints can move sex ratios by limiting the quantity response ([Li et al., 2011](#); [Ebenstein, 2011](#)). We argue that the key distinction is whether fertility can adjust and whose income changes. When fertility is more constrained, as in China, families cannot respond to income gains by having more children, so the gains show up in the sex of the births that remain, which explains why income raised sex selection there. Rising female returns instead raised daughters’ value and the investment each daughter received, which explains why female-specific returns reduced it. In less constrained settings such as Vietnam, fertility can adjust, and the maternal time-cost channel makes the source of income decisive. Empirically, this contribution comes from separating maternal and paternal exposure. Existing studies commonly identify sex-selection effects from geographic or household-level variation ([Qian, 2008](#); [Almond et al., 2019](#); [Anukriti and Kumler, 2019](#); [Anukriti et al., 2022](#); [Chakraborty, 2015](#)). Such shocks assign the same exposure to all households in a place, while our design uses industry-level exposure that varies within the household, which lets the data say

whose income drives selection.

Second, our framework links the fertility and human-capital literatures to shed light on sex selection. Neither margin of the model is new on its own. The maternal time cost of children is a workhorse of family economics (Doepke et al., 2023), and the quantity-quality tradeoff descends from Becker and Lewis (1973) through a long empirical literature (Rosenzweig and Wolpin, 1980; Rosenzweig and Zhang, 2009). What is distinctive is the outcome and the test. We bring these two standard margins to the sex ratio at birth under son preference and confront their joint predictions with parent-specific exposure across sex selection, maternal labor, and fertility. In India, Anukriti et al. (2022) and Jayachandran and Kuziemko (2011) trace how son preference and prenatal selection shape postnatal outcomes, following investments and survival among children already born, while our margin is prenatal. The shock in Anukriti et al. (2022) changes access to sex-selection technology, and its identifying variation, like most in this literature, is shared by every member of the household. It shows how families respond when selection becomes cheaper or fertility more constrained, but not whose labor-market position drives the response. Our shock is a demand-side labor-market change that, combined with parent-specific industry data, allows us to test whether the source of labor-market gains affects fertility and sex composition at birth. The distinction matters for policy. If access drives selection, regulating the technology is the lever. If maternal time and education costs drive it, the levers are likely childcare provision and the cost of schooling.

Third, the paper contributes to the Vietnam BTA literature, which documents formalization, poverty reduction, and firm entry following the trade agreement (McCaig, 2011; McCaig and Pavcnik, 2018; McCaig et al., 2025). Recent work shows that the same shock also affected family and human-capital margins. McCaig et al. (2022) show that the BTA affected youth migration, enrollment, work, and education spending, while Mitra et al. (2026) show that the BTA increased educational attainment and occupational mobility for both sons and daughters, with especially large effects on daughters' college education.² These findings provide empirical context for the quantity-quality channel in our model. Export expansion changed children's human-capital margins, especially for daughters. We build on this evidence but study a different margin and use a different design.

The results suggest that policies aimed only at raising household income are unlikely to break the link between better jobs for mothers and sex selection.³ The model points instead to policies that reduce the maternal time cost of children, such as institutional childcare and related supports. The education-price extension explains how per-child investments can amplify the mechanism,

²Huynh and Ku (2025) show, using province-level shift-share exposure, that the BTA reduced female labor force participation among married and wealthier women. Drozdoff et al. (2025) study how foreign manufacturing jobs in Vietnam delay marriage and childbirth through migration. We differ from both papers in shock, margin, or design, focusing on the BTA tariff shock and on sex composition at birth.

³Anukriti (2018) studies Devi Rupak, a Haryana cash-transfer program rewarding small families, especially those with daughters. Birth rates fell, but the sex ratio at birth worsened, consistent with the fertility-decline channel in our framework. Our paper differs in the source of the shock, a labor-market change that raises maternal opportunity cost, and in our design, which separates maternal from paternal exposure to isolate the source of income.

but it does not imply that lowering education prices alone would necessarily reduce sex selection. Education costs enter both the price of fertility and the relative price of selecting sons. The cleaner policy implication is to loosen the maternal childcare constraint that makes higher female wages translate into lower fertility. The remainder of the paper presents the model, the empirical design, the main estimates, and tests that discriminate among alternative mechanisms.

2. Model

This section formalizes a household model with two margins, the maternal time cost of fertility and quantity-quality investment, and derives discriminating predictions for maternal, paternal, and provincial exposure. The baseline specification develops the fertility margin under maternal childcare time costs and son preference, and derives the shadow prices and comparative statics with respect to the maternal wage, paternal wage, and non-labor income. The [Becker and Lewis \(1973\)](#) quantity-quality margin enters through education investment per child, which adds education costs to the price of fertility while also making daughters more valuable when maternal wages rise. The resulting predictions allow the empirical analysis to distinguish the maternal time-cost mechanism from income and relative-returns explanations for rising sex ratios.

2.1 Baseline Model

Consider a household with pre-existing son preference, rooted in patrilineal kinship norms where sons provide old-age support and continue the family line ([Belanger, 2002](#)). The expected number of boys is $B = bn$, where n is total fertility and $b \in [0, 1]$ is the probability that a birth is male, which the household can influence through sex-selective practices.⁴ The household chooses consumption c , the mother's leisure ℓ , total fertility n , and the male birth probability b to maximize

$$\max_{c, \ell, n, b} U(c, \ell, n, B) \quad \text{where } B = bn, \quad (1)$$

with U increasing and strictly concave in each argument.⁵

Each child requires γ units of the mother's time, which competes with market work and leisure.⁶ The mother allocates her unit time endowment among leisure, market work, and childcare:

$$\ell + h_m + \gamma n = 1, \quad (2)$$

⁴The variable b translates directly to the sex ratio at birth: $\text{Sex Ratio} = 100 \times b / (1 - b)$. For example, $b = 0.52$ corresponds to 108 males per 100 females.

⁵Assume U is twice continuously differentiable and the solution is interior.

⁶In our data, Vietnamese women spend over five hours per day on childcare compared to men's three hours ([ActionAid-Vietnam, 2016](#)). Formal childcare services remain limited outside major urban centers, and social norms assign childcare primarily to mothers. This asymmetry matches the broader time-use regularity emphasized by [Doepke et al. \(2023\)](#). Women bear most childcare time costs, and fertility is lower where childcare burdens fall more disproportionately on women.

where h_m is her labor hours.⁷ The mother earns wage w_m , the father earns wage w_d , and the household receives non-labor income I , such as income from land, transfers, and remittances. Substituting the time constraint (2), the budget constraint is

$$c + w_m \ell + w_m \gamma n + p_b b n = W, \quad (3)$$

where $W = I + w_m + w_d$ is full household income and p_b is the cost per expected boy from sex selection.⁸ The parameter p_b encompasses the monetary costs of sex-determination technology (ultrasound fees, sex-selective abortion) and indirect costs such as health risks to the mother, psychological distress, and social stigma.

The first-order conditions with respect to (c, ℓ, n, b) are derived in Appendix A.1, which also provides the baseline comparative statics:

$$\frac{U_\ell}{U_c} = w_m, \quad \frac{U_n + U_B b}{U_c} = w_m \gamma + p_b b, \quad \frac{U_B n}{U_c} = p_b n. \quad (4)$$

Using the sex-selection condition to simplify the fertility condition gives effective shadow prices $(1, w_m, w_m \gamma, p_b)$ for the four goods (c, ℓ, n, B) . The maternal wage enters two of the four prices. The paternal wage enters none, affecting demands only through full income W .

The price ratio between fertility and boys,

$$\frac{U_n}{U_B} = \frac{w_m \gamma}{p_b}, \quad (5)$$

is the key object for comparative statics. When the maternal wage rises, $w_m \gamma$ increases and children become more expensive relative to expected boys. The fertility-price channel reduces desired fertility and raises the desired boy share $b = B/n$. With son preference, a household that has fewer births has a stronger incentive to concentrate those births on sons. The higher leisure price, $p_\ell = w_m$, reinforces the same substitution away from maternal time-intensive activities. The income effect is weakly positive if the income elasticity of boys is at least as large as that of fertility. Together these forces imply $\partial b / \partial w_m > 0$. For fertility, the substitution effect is unambiguously negative, and when it dominates the income effect, $\partial n / \partial w_m < 0$. Maternal labor supply rises because a higher wage draws the mother into market work and lower fertility frees up childcare time.

The paternal wage w_d operates through income alone, since it enters no effective price. Under the same income-elasticity condition, the income effect on b is weakly positive, but it lacks the substitution force that drives the maternal channel. The model therefore predicts at most a small positive paternal effect on sex selection, operating only through the income effect. The paternal

⁷We normalize paternal labor supply to one, so the father contributes earnings w_d but bears no childcare time cost. This abstraction isolates the maternal time-cost channel and is consistent with evidence that men's labor supply is largely unresponsive to fertility (Angrist and Evans, 1998).

⁸The income-side budget is $c + p_b b n \leq I + w_m(1 - \ell - \gamma n) + w_d$. Rearranging gives the full-income form.

wage raises fertility ($\partial n / \partial w_d > 0$), and non-labor income I operates through the same income-only channel.

The baseline model connects panels (b), (c), and (a) of Figure 1. Rising female wage employment raises the maternal time cost of children, lowers fertility, and increases the male share at birth. The remaining trend is the closing education gender gap in panel (d). The baseline treats each child's cost as maternal time only, but in practice wages also determine how much parents invest in children. The following extension adds this quantity-quality margin.

2.2 Human Capital Investment

The intuition for this extension is as follows. Because daughters' labor market returns also rise with the maternal wage, parents increase investment especially in daughters, narrowing the son-daughter education gap and raising the daughter-education component of child costs. The higher per-child cost reinforces the time-cost pressure to reduce fertility. At the same time, when daughters' investment catches up, replacing a daughter with a son adds less extra education spending than before. Both price channels push toward a higher male birth share. Higher daughter returns also make daughters more valuable, partially offsetting the incentive to select for sons. The asymmetry with the paternal side is equally important. When paternal wages rise, sons' returns increase and parents invest more in sons. The resulting son-daughter investment gap raises the cost of selecting a son relative to a daughter, offsetting the income and son-value forces that would otherwise raise sex selection.

Parental wages serve a dual role. Beyond determining time costs, they signal the labor-market returns to children's human capital. Children's labor-market outcomes correlate with same-sex parental earnings, so maternal wages proxy for the return on daughters' education and paternal wages proxy for the return on sons' education. We formalize this dual role by setting $r_g = w_m$ and $r_b = w_d$, where r_g and r_b denote the returns to girls' and boys' human capital.⁹ This extension embeds our model in the quantity-quality framework of Becker and Lewis (1973).

Parents invest e_b in each son and e_g in each daughter, where e measures educational input (such as years of schooling). A human capital production function $h(e)$ converts educational input into effective units of human capital, with $h' > 0$, $h'' < 0$, and $h(0) = 0$. The market returns r_b and r_g price each efficiency unit, so a child's human capital value is

$$H_b = r_b \cdot h(e_b), \quad H_g = r_g \cdot h(e_g). \quad (6)$$

Parents value children's human capital through $V(\bar{E})$ appended to the baseline utility, where $\bar{E} = b r_b h(e_b) + (1 - b) r_g h(e_g)$ is the expected per-child human capital and V is increasing and

⁹The mapping $r_g = w_m$ is most natural for individual-level variation, where a mother's own industry determines both her wage and the labor market signal for daughters' future earnings. At the province level, returns may differ from individual wages when local industry composition shifts daughters' future prospects independently of the mother's own wage. Appendix A.2.1 generalizes the model to allow returns to differ from individual wages, and Section 7 tests this empirically using female-weighted provincial exposure P_p^w .

concave. The household now chooses $(c, \ell, n, b, e_b, e_g)$ jointly to maximize

$$U(c, \ell, n, B) + V(\bar{E}) \quad (7)$$

subject to the budget constraint

$$c + w_m \ell + n[w_m \gamma + p_e(b e_b + (1 - b) e_g)] + p_b b n = W, \quad (8)$$

where p_e is the unit cost of educational input (tuition, school fees). The key feature of this budget constraint is that education costs scale with both the number of children and investment per child. Adding one more child means paying for that child's education at the chosen quality level. Raising education investment means paying the additional schooling cost for every child. Each margin raises the cost of the other, so parents trade off between quantity and quality.

Sex composition matters because sons and daughters may require different education spending. When $r_b > r_g$, sons receive more investment than daughters, so replacing a daughter with a son raises costs by $p_e(e_b - e_g)$. Fertility and quality are also linked. When fertility falls, parents invest more in each remaining child, raising the marginal cost of another child. This quantity-quality tradeoff connects education investment to both fertility and sex selection.

The first-order conditions for e_b and e_g (Appendix A.2) equate the marginal return per dollar across genders:

$$r_b h'(e_b^*) = r_g h'(e_g^*). \quad (9)$$

Because h is concave ($h'' < 0$), the gender with higher market returns receives more investment. When $r_b > r_g$, $e_b^* > e_g^*$. When maternal wages rise relative to paternal wages, daughters' returns increase and the investment gap narrows, matching the closing education gender gap in Figure 1, panel (d).

Grouping terms by n and $B = bn$ decomposes the budget into a fertility component and a sex-selection component:

$$c + w_m \ell + n[w_m \gamma + p_e e_g] + B[p_b + p_e(e_b - e_g)] = W. \quad (10)$$

The fertility price $p_n = w_m \gamma + p_e e_g$ now includes both childcare time and daughter education costs. The sex-selection price $p_B = p_b + p_e(e_b - e_g)$ reflects the incremental cost of a son relative to a daughter. When $w_d > w_m$, sons receive more investment than daughters ($e_b^* > e_g^*$), so the education gap $p_e(e_b - e_g) > 0$ adds to the cost of sex selection beyond p_b . Reducing γ directly weakens the maternal time-cost channel. Changing p_e is not an equivalent policy lever, because p_e enters both p_n and p_B and the two movements work against each other. Appendix A.2 provides further discussion of this asymmetry.

Consider the effect of a rise in w_m on sex selection. The human capital extension introduces two channels that work in opposite directions. A cost channel operates through the effective prices.

Higher $r_g = w_m$ raises daughters' education investment, which increases the fertility price p_n and, when the education gap narrows, lowers the sex-selection price p_B . Both movements amplify the baseline time-cost effect. By contrast, a value channel operates through $V(\bar{E})$. Higher daughters' earnings make them more valuable to the household, reducing the incentive to select for sons.

The cost channel pushes b up through multiple reinforcing price effects. The higher leisure price, the amplified fertility price, and the lower education-cost penalty for selecting sons when daughters' investment catches up. The income effect reinforces the same direction, while only the value channel pushes b down. The net maternal effect on sex selection is likely positive.

This extension also changes the paternal wage prediction. Higher w_d raises sons' returns ($r_b = w_d$), making sons more expensive to educate. The higher cost of educating sons raises the sex-selection price p_B and offsets the income and son-value forces that push toward more sex selection. Because w_d does not raise the maternal time price of children, these opposing forces can cancel, yielding an ambiguous and likely small paternal effect on sex selection. For fertility, the leading paternal force remains the positive income effect. Any increase in p_n from higher desired child quality is indirect and second-order. Non-labor income I enters only through full income W without affecting education returns, leaving only the income effect.

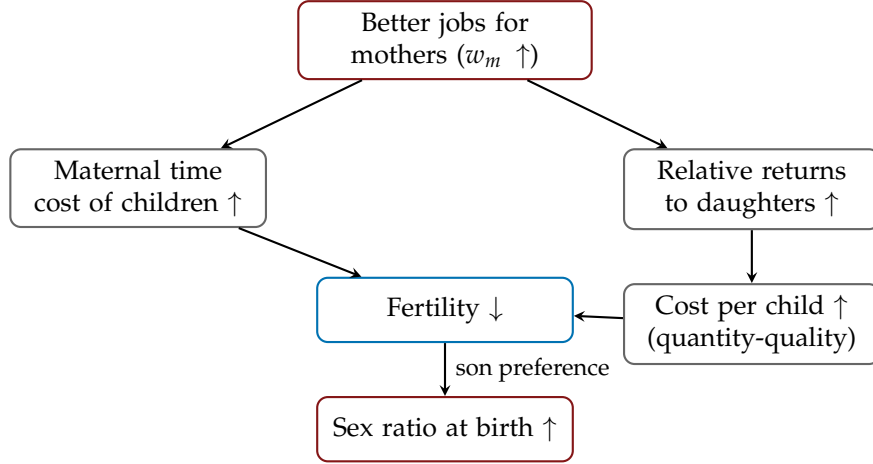
Thus the extension adds panel (d) of Figure 1 to the baseline account. Maternal wage gains can narrow the education gender gap while still reinforcing the pressure toward fewer births and a more male-skewed birth ratio.

Figure 2 assembles the model in one picture. A maternal wage gain enters the household problem twice. It raises the value of the mother's time, which prices every additional birth through the childcare she supplies, and it raises the returns to daughters, which raises desired investment per child and with it the cost of each birth through the quantity-quality margin. Both paths lower planned fertility, which microfound the fertility-decline account of Jayachandran (2017) with two classic margins, the maternal time cost of the fertility literature (Doepke et al., 2023) and the quantity-quality tradeoff of Becker and Lewis (1973). Under son preference, fewer planned births leave fewer chances to obtain a son, and selection rises. What the model adds is the assembly. Each link is a standard force from the fertility, quantity-quality, and sex-selection literatures, but placing them in one household problem shows that they share a single trigger, the mother's returns on the labor market, and a single bottleneck, fertility. Income arriving through the father or through non-labor sources enters only the budget, so it raises fertility while leaving the mother's time price and the selection margin unchanged. The human-capital extension adds one offsetting force for fathers, whose higher wages also raise the cost of educating sons.

Table 1 collects the comparative statics. The rows distinguish maternal wages, paternal wages, and non-labor income, while the columns show the predicted signs for sex selection, maternal labor supply, and fertility. We highlight three predictions that organize the empirical analysis.

Prediction 1 (Maternal time cost). *Holding w_d and I fixed, an increase in w_m raises the male birth share, $\partial b / \partial w_m > 0$, raises maternal labor supply, $\partial h_m / \partial w_m > 0$, and lowers fertility, $\partial n / \partial w_m < 0$.*

Figure 2: Model Overview



Notes: Boxes are model objects and arrows are the model's causal channels, derived in Section 2. Red marks the maternal wage origin and the sex-ratio outcome.

Prediction 2 (Paternal income). *Holding w_m and I fixed, an increase in w_d raises fertility, $\partial n / \partial w_d > 0$, does not raise maternal labor supply, $\partial h_m / \partial w_d \leq 0$, and leaves the male birth share approximately unchanged, $\partial b / \partial w_d \approx 0$.*

Prediction 3 (Non-labor income). *Holding w_m and w_d fixed, an increase in I raises fertility, $\partial n / \partial I > 0$, weakly lowers maternal labor supply, $\partial h_m / \partial I \leq 0$, and leaves the male birth share approximately unchanged, $\partial b / \partial I \approx 0$.*

Each prediction, taken alone, admits rival explanations. A positive maternal coefficient on male births is consistent with a pure income story, and a fertility decline is consistent with many forces. The discriminating content is the joint pattern. Only a shock to the mother's own wage produces more work, fewer births, and more sons at once, while the same shock arriving through the father or through non-labor income raises births without moving the birth composition. Each empirical section tests one column of Table 1 and refers back to Predictions 1 through 3. Section 7 then asks whether any rival explanation reproduces the full pattern.

These predictions apply to settings like Vietnam where fertility can adjust. Under binding fertility constraints, as under China's one-child policy (Zhang, 2017), the model generates different predictions (Appendix A.2.2 provides the details). When the fertility margin is constrained, the fertility-price channel is shut down, so the marginal maternal-wage effect works through leisure prices, sex-selection prices, income, and the value of daughters rather than through a reduction in n . Separately, income gains that have only income-effect responses in our setting may become positive predictors of sex selection, because the composition of births becomes the main remaining adjustment margin. At the same time, high per-child investment (a consequence of the binding quantity constraint) amplifies the value channel. When female returns rise, daughters become sub-

Table 1: Model Predictions

Income source	Outcome		
	(1)	(2)	(3)
	Sex selection (male share b)	Maternal labor (hours h_m)	Fertility (births n)
Mother’s wage (w_m)	+	+	–
Father’s wage (w_d)	0	–	+
Non-labor income (I)	0	–	+

Notes: + positive, – negative, 0 small or negligible predicted effect. Predictions summarize the model’s key comparative statics in Section 2. Appendix A provides the derivations.

stantially more valuable because each daughter receives heavy investment. The relative-returns channel, which is dominated by time costs in Vietnam, gains force when the fertility margin is shut down. Thus the same framework highlights maternal time costs in Vietnam, while also explaining why income shocks and female returns can have stronger composition effects in China’s constrained-fertility setting (Almond et al., 2019; Qian, 2008).

3. Data and Measures

We bring the theoretical framework to the data using Vietnam’s 2001 Bilateral Trade Agreement (BTA) shock. This section describes the institutional environment for sex selection in Vietnam, constructs three exposure measures corresponding to the maternal-wage, paternal-wage, and broad local-income channels summarized in Table 1, then describes the household survey and census data used to observe parents’ industries and birth outcomes.

3.1 Institutional Context for Sex Selection

Son preference in Vietnam is rooted in patrilineal kinship. Sons are believed to continue the family line, perform ancestor worship, and provide old-age support, obligations that daughters transfer to their husbands’ families at marriage (Belanger, 2002; Guilmoto, 2012). Fertility itself was already low at the start of our study period, near two births per woman (panel (c) of Figure 1). Vietnam’s total fertility rate reached replacement around 2000 under a longstanding one-or-two-child policy norm, so parents who wanted a son faced the choice within a small number of births rather than through continued childbearing (Guilmoto, 2012).

The technology for prenatal sex selection was in place before the BTA’s effects could appear in births. Ultrasound diagnosis diffused rapidly through public and private clinics from the late 1990s, and abortion has long been legal, inexpensive, and widely provided through the public family-planning system, giving Vietnam one of the highest abortion rates on record (Teer-

awichitchainan and Amin, 2010; Guilmoto, 2012). Ultrasound followed by induced abortion is the dominant prenatal channel, with pre-conception timing methods documented but secondary (Guilmoto, 2012). The 2003 Population Ordinance and its implementing decree prohibited fetal sex disclosure and sex-selective abortion, but enforcement was limited, and the national sex ratio at birth began its sustained rise around 2003, reaching 112 to 113 boys per 100 girls by the mid-2010s with the strongest imbalances in the northern regions (Becquet and Guilmoto, 2018).

3.2 The 2001 BTA and Exposure Measures

Before the BTA went into effect on December 10, 2001, Vietnamese exports to the US faced high tariffs under the Column 2 schedule applied to non-market economies. The BTA granted Vietnam Most Favored Nation (MFN) status, dramatically reducing US tariffs on Vietnamese exports while having minimal impact on Vietnam’s import tariffs.¹⁰ Following the agreement, Vietnam’s exports to the US surged from 0.7 billion dollars in 2000 to nearly 10 billion dollars in 2008.

We construct two complementary formulas, yielding three exposure measures, to capture how these differential tariff cuts transmit to fertility and sex selection through the labor market channels described in the model. The first captures direct exposure through parents’ industries:

$$\tau_j = \log(1 + \text{Column } 2_j) - \log(1 + \text{MFN}_j), \quad (11)$$

where j denotes a parent’s industry. A higher value of τ_j indicates a larger effective industry tariff cut. Because mothers and fathers tend to work in different industries, we construct separate measures for each parent to capture relative changes in labor market returns.¹¹ These industry-level measures proxy for the maternal and paternal wage channels (w_m and w_d) in Table 1. Prior work using the same tariff variation shows that the BTA generated meaningful labor-market adjustment. McCaig and Pavcnik (2018) document large-scale reallocation from informal to formal employment, and McCaig et al. (2025) show growth in foreign and domestic firm entry in more exposed industries. We confirm this formalization channel directly in our own sample of parents at the time of fertility decisions below.

The second measure is a province-level shift-share that captures broader local economic effects:

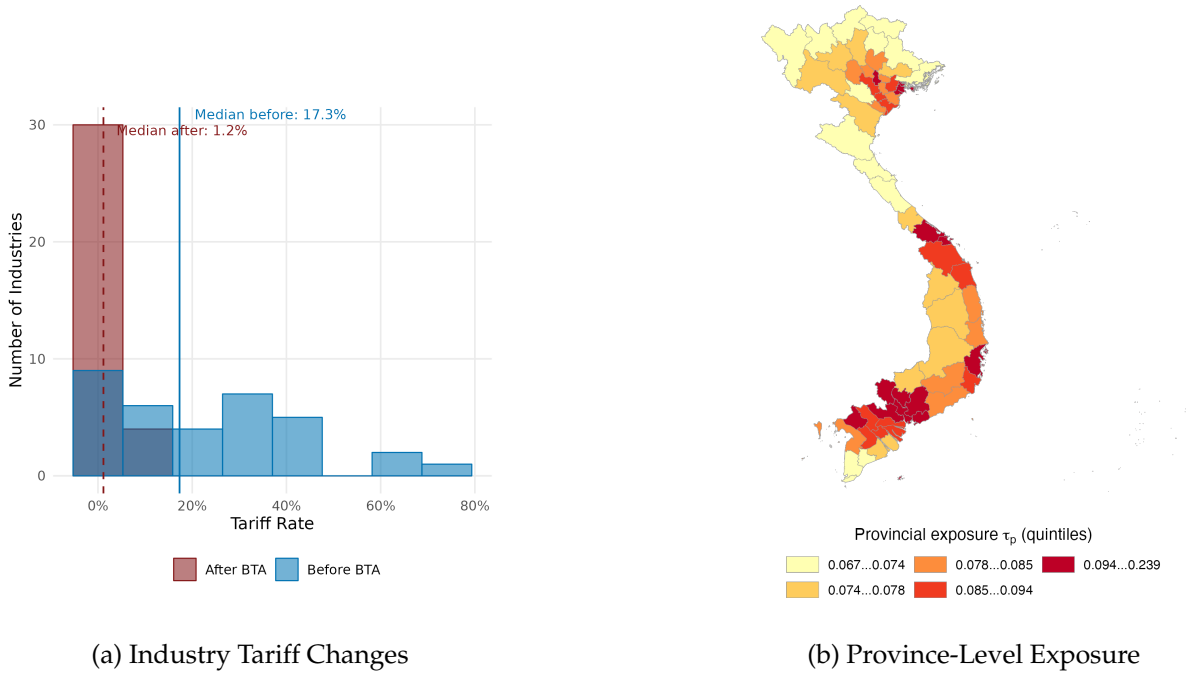
$$\tau_p = \sum_j s_{jp} \tau_j, \quad (12)$$

where s_{jp} is the share of province p ’s employment in tradeable industry j in the 1999 Census. Following Kovak (2013), employment shares are computed over all tradeable industries. This

¹⁰While Vietnam was obligated to reduce tariffs mainly on agricultural and food products, these reductions were minimal, ranging from 0.03 to 2.7 percentage points. In addition to tariffs, Vietnam agreed to eliminate various import quotas. Most of these quotas had, however, been eliminated by the end of 2002 (STAR-Vietnam, 2003).

¹¹The correlation between maternal and paternal tariff exposure in our sample is consistently low, ranging from 0.24 to 0.36 across household survey years, described below, confirming that parents face different levels of exposure. The correlation between individual parental exposure and the provincial shift-share τ_p is even lower, remaining below 0.15 throughout the sample period. Appendix Figure C1 displays the full pairwise correlation structure by wave.

Figure 3: Variation in Trade Exposure Across Industries and Provinces



Notes: Panel (a) shows US tariff rates on Vietnamese exports before (Column 2) and after (MFN) the BTA across 2-digit ISIC industries, from [McCaig \(2011\)](#). Panel (b) shades provinces by quintiles of the shift-share exposure $\tau_p = \sum_j s_{jp} \tau_j$, where s_{jp} is industry j 's employment share in province p (1999 Census). Darker shading indicates greater exposure.

provincial measure captures broad local economic effects rather than any individual household's non-labor income. It proxies for the I channel in Table 1, but at the province level. Export expansion can raise returns to land and agricultural assets, and income gains can spread through transfers within extended families. Consistent with this broad local-income interpretation, [McCaig \(2011\)](#) use this province-level exposure measure to show that the BTA significantly reduced poverty rates across Vietnamese provinces. The provincial measure τ_p therefore captures reduced-form local gains that may flow through household and community networks rather than through a parent's own industry alone.

Figure 3 illustrates the variation in both measures. Panel (a) shows the distribution of US tariff rates before and after the BTA across two-digit industries. The tariff reductions were large. The median Column 2 rate was 17.3 percentage points, falling to a median MFN rate of just 1.2 percentage points, a near-complete elimination of tariff protection. In log-difference terms, the industry tariff cut τ_j has a median of 0.168, reaching a maximum of 0.438 in the most exposed sector. A parent in the most exposed industry faced a tariff reduction roughly 2.6 times the median, implying substantially different incentives for formal employment and wage growth across the treatment distribution. Industry codes classify workers by activity regardless of formality, so informal-sector

workers in exposed industries count as exposed on the same basis as formal workers. Only about one in four working mothers held a wage job before the BTA, so the scope for movement into wage work in exposed industries was large.

Panel (b) maps the resulting province-level exposure. The measure varies by a factor of 3.6 across 61 provinces, from 0.067 in the least exposed to 0.239 in the most exposed (mean 0.089, median 0.081). Above-median exposure is concentrated in southern provinces around Ho Chi Minh City and along the central and southern coast, areas with established export infrastructure and a higher pre-existing share of employment in manufacturing. Northern inland and mountainous provinces, where employment was concentrated in agriculture, show the lowest exposure.

For the main analysis, we define $\tau_{m(i)}$ and $\tau_{d(i)}$ as the tariff cuts of the mother’s and father’s industries, and $\tau_{p(i)}$ as the province-level shift-share exposure.¹² Because the two-way fixed effects coefficient on a continuous treatment averages dose-specific effects (Callaway et al., 2024), we report parent coefficients per 0.10 increase in exposure and province coefficients per one standard deviation. Because τ_j is a log difference of gross tariff rates, a 0.10 increase in exposure corresponds approximately to a 10 percentage-point larger tariff cut. Section 6 reports binary high-versus-low exposure contrasts as robustness checks, together with diagnostics for the continuous specification.

3.3 Household Surveys and Census Data

To study how trade exposure affects sex ratio at birth, the ideal dataset would record each child’s sex alongside both parents’ industry, employment, and location around conception, allowing us to assign the trade shock separately to mothers, fathers, and provinces. The sex-selection literature typically relies on Demographic and Health Surveys (DHS) or population censuses, but DHS data rarely measure fathers’ labor-market exposure, while infrequent censuses can observe parents’ industry and location only years after conception.¹³

We therefore use the Vietnam Household Living Standards Surveys (VHLSS) (General Statistics Office of Vietnam, 2010), a biennial nationally representative survey spanning 2002 through 2010 that records industry, location, and employment for both parents at each survey wave. Because the VHLSS does not contain complete birth histories, we work with the flow of births in each wave, treating each cross-section as a sample of infants aged 0–1 observed alive and co-resident with their parents. This choice trades birth-history completeness for contemporaneous measurement of both parents’ industries, which is essential for separating maternal and paternal exposure. We supplement the VHLSS with the 1999 Population Census from IPUMS International (Ruggles et al., 2025), whose province-level employment shares serve as weights for the shift-share

¹²Roughly 21 percent of parents work in industries where the Column 2 and MFN rates are both zero, yielding $\tau_j = 0$. These workers are concentrated in non-export service and agricultural sectors.

¹³DHS surveys provide birth histories, but they contain limited information on fathers and are conducted only intermittently. Population censuses are even more infrequent and come with a different problem. Treatment must be assigned using industry and location at the census date rather than at conception. If workers switch industries or move after the shock, treatment is mismeasured and estimated effects are attenuated toward zero.

exposure measure and whose birth records extend the pre-treatment baseline.

The Census provides births from 1997–1999, contributing approximately 49,000 infant observations to the pre-treatment baseline. The VHLSS waves from 2002 through 2010 contribute repeated cross-sections of infants aged 0–1, yielding approximately 17,700 infant-parent observations across five waves. For each infant we observe the child’s sex and the parents’ contemporaneous industry codes. In the VHLSS, we also observe parental work hours and job types. The Census therefore contributes to the sex selection analysis, while the labor supply and fertility analyses use VHLSS measures described below.

Exposure assignment requires a reported industry for each parent. The estimation sample therefore consists of births for which both the mother and the father report an industry code, and mothers without market work are excluded because no industry exposure can be assigned to them. This restriction matches the mechanism under study. Female labor force participation in Vietnam was already around 70 percent before the BTA, so the relevant margin is not entry into work but the quality of work available to working mothers, and the wage-employment outcome accordingly measures movement between wage jobs and non-wage work such as farm or household-enterprise activity among mothers with jobs.

A key measurement challenge in this setting is identifying parent-child relationships within extended family households, where multiple generations often live together and survey relationships are recorded relative to the household head rather than within nuclear families. We develop an algorithm that identifies both couple and parent-child relationships within extended households, validated against known parent-child links in the 2014 and 2016 VHLSS waves, achieving 93 percent accuracy for mothers and 91 percent for fathers. Appendix B details this procedure. Roughly 35 percent of the VHLSS sample lives in extended households, so a substantial share of infants is linked through subfamilies other than the household head’s, and the validation exercise covers precisely these harder cases.

Table 2 presents summary statistics for Census 1999 and the five VHLSS waves. Mothers’ baseline characteristics are comparable across sources, with average ages around 27–28 years and education ranging from 6.5 to 7.4 years across waves. The proportion of urban residents rises over time, consistent with Vietnam’s urbanization trend. The male birth share is higher in the VHLSS waves than in the 1997–1999 Census baseline. The Census implies a sex ratio at birth of about 107 boys per 100 girls, while the 2002–2008 VHLSS waves imply ratios between 110 and 114. The estimation samples therefore capture the same broad upward shift documented in Figure 1.

The model predicts that maternal and paternal wages have opposing effects on fertility (Table 1, column 3). Our main analysis conditions on the birth of an infant aged 0–1, so fertility, whether to have a birth at all, must be studied in a separate sample that includes woman-years without births. In the absence of complete birth histories, a common data challenge in this literature, we reconstruct annual birth outcomes from the household roster. Because the roster records the birth years of all children currently living with the mother, we can infer in which years she

Table 2: Summary Statistics for Census 1999 and VHLSS Survey Waves

	Census	VHLSS				
	1999	2002	2004	2006	2008	2010
Male birth	0.518 (0.500)	0.532 (0.499)	0.529 (0.499)	0.524 (0.499)	0.527 (0.499)	0.511 (0.500)
Mother's age	27.00 (5.05)	28.10 (5.19)	28.39 (5.29)	27.88 (5.27)	28.09 (5.13)	28.24 (5.04)
Mother's educ. (yrs)	6.46 (3.43)	6.72 (3.60)	7.17 (3.71)	7.42 (3.69)	8.01 (3.71)	8.22 (3.66)
Ethnic minority	0.201 (0.401)	0.206 (0.404)	0.218 (0.413)	0.215 (0.411)	0.204 (0.403)	0.213 (0.409)
Urban	0.131 (0.337)	0.175 (0.380)	0.211 (0.408)	0.231 (0.421)	0.266 (0.442)	0.270 (0.444)
Birth order	1.93 (0.81)	2.07 (0.80)	2.02 (0.80)	1.90 (0.80)	1.88 (0.78)	1.89 (0.76)
Mother wage earner	—	0.202 (0.402)	0.243 (0.429)	0.275 (0.446)	0.333 (0.471)	0.366 (0.482)
Father wage earner	—	0.372 (0.483)	0.397 (0.489)	0.420 (0.494)	0.486 (0.500)	0.512 (0.500)
<i>N</i>	48,897	5,934	3,159	3,034	3,164	2,374

Notes: Standard deviations in parentheses. Census 1999 covers 1997–1999 births. VHLSS waves include children aged 0–1. Dashes indicate variables unavailable in the Census.

had a birth. For each woman in a given survey year, we create one observation per year from when she turned 20 until the survey date, coding the outcome as one if a child in the roster has the corresponding birth year. This reconstruction yields approximately two million woman-year observations for the fertility analysis.

The main limitation of the retrospective measure is that children who have left the household through marriage, migration, non-co-residence, or death are absent from the roster, so earlier births are systematically missing for older women. The measure also cannot observe fetal losses. We therefore interpret the fertility analysis as an annual birth-hazard test of whether the same exposure variation changes the probability of a birth in a given year. Because the roster does not record marriage dates, the fertility measure also cannot distinguish births postponed by later marriage from fewer births within marriage.

For the individual exposure measures, migration across provinces is not the direct source of

treatment mismeasurement because maternal and paternal exposure are assigned from each parent’s industry, not location. The main concern is industry switching, which Section 4 discusses in detail. To address this selection problem, we exploit the VHLSS’s rotating panel feature, which tracks a subset of households across three consecutive waves (2002–2004–2006). From these households we draw women aged 20–40 as our analysis sample. By fixing each parent’s industry affiliation at the 2002 wave, the earliest available survey and only months after the December 2001 implementation, we limit contamination from compositional changes driven by post-shock worker reallocation. The panel structure also allows us to control for time-invariant unobservables, including fertility preferences and son preference, through individual fixed effects. Given the short panel length, we rely on the cross-sectional variation for our main analysis and use the panel estimates as a robustness check.

Appendix Table C1 compares the 2002 wave of the rotating-panel sex-selection sample to the main 2002 cross-section on baseline demographics. Mothers’ age, education, minority status, urban share, and province-level tariff exposure are closely matched across the two samples, suggesting that the rotating-panel entrants are broadly similar to the main cross-section on observables.

4. Empirical Strategy

Testing the comparative statics in Table 1 requires an empirical design that separates direct maternal exposure, direct paternal exposure, and broader local exposure from a single trade shock. We embed the exposure measures τ_m , τ_d , and τ_p defined in Section 3.2 in a difference-in-differences framework that exploits industry-level tariff cuts and the province-level exposure induced by pre-BTA industry composition. Estimating the three channels jointly lets us distinguish the maternal labor-market mechanism from paternal-income and broader local-income channels within the same trade shock.

The pre- and post-treatment periods differ across outcomes. For sex selection and fertility, we set $\text{Post}_t = \mathbf{1}[\text{birthyear} > 2001]$, so that Census births from 1997–1999 are entirely pre-treatment while children born in 2002 or later are post-treatment. Since the BTA took effect in December 2001, the earliest post-treatment birth cohorts include children conceived before the agreement and are therefore only partially exposed. For maternal labor supply, we set $\text{Post}_s = \mathbf{1}[\text{survey year} > 2002]$, making the 2002 VHLSS the pre-treatment baseline and waves from 2004 onward post-treatment.

Our baseline specification for sex selection estimates a linear probability model where the unit of observation is a child:

$$y_{it} = \beta_1(\tau_{m(i)} \times \text{Post}_t) + \beta_2(\tau_{d(i)} \times \text{Post}_t) + \beta_3(\tau_{p(i)} \times \text{Post}_t) + \alpha_t + \delta_{p(i)} + \gamma_{j_{m(i)}} + \gamma_{j_{d(i)}} + X'_{it}\phi + \varepsilon_{it}. \quad (13)$$

Here y_{it} is an indicator equal to one if child i born in year t is male, with $p(i)$, $m(i)$, and $d(i)$ denoting child i ’s province of birth, mother, and father, respectively. The exposures $\tau_{m(i)}$, $\tau_{d(i)}$, and $\tau_{p(i)}$ are assigned through the mother’s industry, the father’s industry, and the province of birth,

respectively. In later sections we write τ_m , τ_d , and τ_p without unit subscripts when the reference is unambiguous. The specification includes birth-year fixed effects α_t , province fixed effects $\delta_{p(i)}$, and ISIC industry fixed effects for the mother's industry $\gamma_{j_{m(i)}}$ and the father's industry $\gamma_{j_{d(i)}}$.

The vector X'_{it} includes the mother's education, urban residence, and minority status. It also includes world-average tariff controls at the parents' 2-digit ISIC industries matched by birth year, which absorb other industry-level tariff movements during the study period, including contemporaneous multilateral liberalization around Vietnam's WTO accession in January 2007. Interactions of pre-BTA provincial employment shares in agriculture and manufacturing with year fixed effects absorb differential trends arising from preexisting sectoral composition. Standard errors are clustered at the province level (61 clusters). We cluster at the province level as the baseline because the VHLSS samples communes within provinces, so outcomes are dependent within provinces by survey design, and the treatment varies along three dimensions, the mother's industry, the father's industry, and the province (Abadie et al., 2023). Nonetheless, Section 6 reports robustness to clustering at the industry level and two-way clustering.

For maternal labor market outcomes, the unit of observation shifts to a mother-survey wave. We estimate by OLS:

$$y_{ms} = \beta_1(\tau_m \times \text{Post}_s) + \beta_2(\tau_{d(m)} \times \text{Post}_s) + \beta_3(\tau_{p(m)} \times \text{Post}_s) + \alpha_s + \delta_{p(m)} + \gamma_{j_m} + \gamma_{j_{d(m)}} + X'_{ms}\phi + \varepsilon_{ms}, \quad (14)$$

where y_{ms} is a labor market outcome for mother m observed in survey wave s , with $p(m)$ her province of residence and $d(m)$ her spouse, and $\text{Post}_s = \mathbf{1}[\text{survey year} > 2002]$. All remaining controls, fixed effects, and clustering follow Equation (13), with tariff controls matched by survey year rather than birth year. The coefficients β_1 , β_2 , and β_3 are effects per unit of the corresponding continuous exposure.

For fertility, the specification replaces province and industry fixed effects with mother fixed effects. Using the retrospective birth panel described in Section 3.3, we define AnyBirth_{mt} as an indicator equal to one if mother m had a birth in year t :

$$\text{AnyBirth}_{mt} = \beta_1(\tau_m \times \text{Post}_t) + \beta_2(\tau_{d(m)} \times \text{Post}_t) + \beta_3(\tau_{p(m)} \times \text{Post}_t) + \alpha_t + \mu_m + X'_{mt}\phi + \varepsilon_{mt}. \quad (15)$$

Mother fixed effects μ_m absorb all time-invariant heterogeneity, including underlying son preference, desired fertility, and permanent differences in industry and location. The control vector X'_{mt} includes interactions of pre-BTA provincial employment shares in agriculture and manufacturing with year fixed effects, and world-average tariff controls matched by birth year. Identification comes from within-mother variation in birth propensity before and after the BTA, comparing mothers with different levels of exposure.

Two threats to the cross-sectional estimates motivate the panel specification. The primary concern is industry switching. The cross-section records each parent's current industry rather than pre-shock affiliation, so workers who move into high-tariff-cut industries after the BTA can con-

taminate $\tau_{m(i)}$ and $\tau_{d(i)}$ with post-shock selection. This concern is most relevant for the maternal and paternal industry coefficients, because treatment is assigned from each parent’s own industry. Migration across provinces is a secondary concern for the province-level coefficient. If the trade shock draws households with stronger son preference into high-exposure provinces, the composition of highly exposed provinces changes in a way that biases β_3 upward. Including τ_p alongside the parent-specific measures helps separate this broader province-level channel from maternal and paternal industry exposure, but it does not address sorting across industries.

To address these concerns, we exploit the VHLSS rotating panel described in Section 3.3 and lock each parent’s treatment status to her pre-BTA industry affiliation, following [McCaig and Pavcnik \(2018\)](#):

$$y_{it} = \beta_1(\tau_{m(i)} \times \text{Post}_t) + \beta_2(\tau_{d(i)} \times \text{Post}_t) + \beta_3(\tau_{p(i)} \times \text{Post}_t) + \alpha_t + \mu_{m(i)} + X'_{it}\phi + \varepsilon_{it}. \quad (16)$$

The unit of observation remains a child i , as in Equation (13), but mother fixed effects $\mu_{m(i)}$ replace province and industry fixed effects, absorbing all time-invariant heterogeneity of the mother $m(i)$ of child i . Parent exposures are assigned based on each parent’s industry reported in the 2002 VHLSS wave, the earliest available survey, and are held fixed across the rotating panel. Since the 2002 wave was fielded shortly after the BTA took effect, exposure is assigned early-post, which limits later industry switching but cannot rule out sorting before the first interview.

The panel sample consists of women aged 20–40 observed in the VHLSS rotating panel. For sex selection, the analysis uses mother-wave observations with eligible recent births and fixes treatment to each parent’s 2002 industry. Identification with mother fixed effects comes from comparing births to the same mother before and after the BTA when such within-mother variation is available. If endogenous sorting were inflating the cross-sectional estimates, fixing treatment to pre-shock industries should attenuate the panel coefficients relative to the cross-sectional results.

The key identifying assumption is that, absent the BTA, outcomes would have evolved similarly across high- and low-exposure groups. The 1999 Census births (1997–1999) and VHLSS births in 2000–2001 provide pre-treatment observations spanning multiple years before the shock, which we use to assess differential pre-trends in Section 5. For the panel specification, fixing treatment to the first VHLSS wave reduces the concern that later industry switching drives exposure, but it still requires that initial industry affiliation was not chosen in anticipation of the BTA’s differential tariff effects across sectors.

The coefficients β_1 , β_2 , and β_3 map directly to the net comparative statics in Table 1. The model predicts $\beta_1 > 0$ for sex selection and maternal labor supply and $\beta_1 < 0$ for fertility. Maternal exposure raises the opportunity cost of childbearing, while the quantity-quality margin reinforces the fertility and sex-selection responses. Paternal exposure (β_2) and province-level exposure (β_3) provide comparison channels because they shift household resources without the same maternal-time component.

5. Main Results

5.1 Effects on Sex Selection

Prediction 1 implies that sex selection should respond specifically to the mother’s own labor-market exposure. Maternal exposure should raise male birth probability because it changes the maternal time cost of children and the quantity–quality margin, while paternal and province-level exposure should have little or no effect on sex selection. Table 3 reports progressively richer specifications of Equation (13), where the dependent variable is an indicator equal to one if the child is male.

Columns (1) and (2) introduce the maternal and paternal exposure channels individually, while column (3) combines them. Columns (4) and (5) add the province-level channel, with column (5) adding the full control set. When entered alone in column (1), a 0.10 increase in maternal industry exposure ($\tau_m \times \text{Post}_t$) raises male birth probability by 2.2 percentage points (s.e. 0.9 percentage points), significant at the 5 percent level.

The father’s industry exposure ($\tau_d \times \text{Post}_t$) enters with a positive but imprecisely estimated coefficient of 0.010 in column (2). In the combined specifications of columns (3) through (5), the paternal coefficient ranges from 0.005 to 0.010 and remains statistically indistinguishable from zero, although the paternal confidence interval is wide and a formal test does not reject equality of the two parent coefficients. The model’s implications concern the sign and size of each channel rather than the difference between them, and the estimates deliver the predicted configuration, a positive and precise maternal response alongside a paternal response near zero (Prediction 2).

Similarly, province-level exposure ($\tau_p \times \text{Post}_t$) is near zero and insignificant throughout, entering at -0.003 in column (4) and 0.002 in column (5) per one-standard-deviation increase. This result matches the comparison channel in Prediction 3. Broad local income gains do not directly raise the mother’s opportunity cost of childbearing, so they should have weaker effects on sex selection than direct maternal industry exposure.

Across the progressive specifications, the maternal coefficient remains significant at the 5 percent level throughout. For a mother whose industry received the sample median tariff cut of approximately 7 percentage points, the estimates in Table 3 imply an increase in male birth probability of about 2 percentage points (1.5 to 2.6 across control sets), equivalent to moving the sex ratio at birth from the unexposed mothers’ 105 to roughly 114 boys per 100 girls.

Anukriti (2018) finds that first births become 0.5 to 1.2 percentage points more likely to be male after the Devi Rupak fertility incentive in India, and Almond et al. (2019) estimate a cumulative increase of approximately 2.8 percentage points over four years for second births following a firstborn girl in China. Our median-cut estimate of 1.5 to 2.6 percentage points sits above the range documented by Anukriti (2018) and below the benchmark from Almond et al. (2019), consistent with the fact that the latter conditions on the subgroup facing the strongest son preference pressure while our estimate pools across all birth orders and sibling compositions. Nonetheless,

Table 3: Effects of BTA Trade Exposure on Male Birth Probability

	MaleBirth				
	(1)	(2)	(3)	(4)	(5)
$\tau_m \times \text{Post}$	0.022** (0.009)		0.021** (0.009)	0.021** (0.009)	0.037** (0.014)
$\tau_d \times \text{Post}$		0.010 (0.013)	0.005 (0.013)	0.005 (0.013)	0.010 (0.013)
$\tau_p \times \text{Post}$				-0.003 (0.008)	0.002 (0.012)
R^2	0.015	0.014	0.015	0.015	0.016
Observations	66,562	66,562	66,562	66,562	64,170
Dep. Var. Mean	0.518	0.518	0.518	0.518	0.518
Controls					✓

Notes: Dep. var. is MaleBirth. Entries are effects of a 0.10 increase in parent-specific exposure. Province exposure is standardized so that its coefficient is the effect of a one-standard-deviation increase. All columns include birth-year, province, and parents' 2-digit ISIC fixed effects. Controls (col. 5) add mother's education, urban residence, minority status, world-average tariff controls at parents' industries, and pre-BTA provincial employment shares $\times$ birth-year. Province-clustered standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

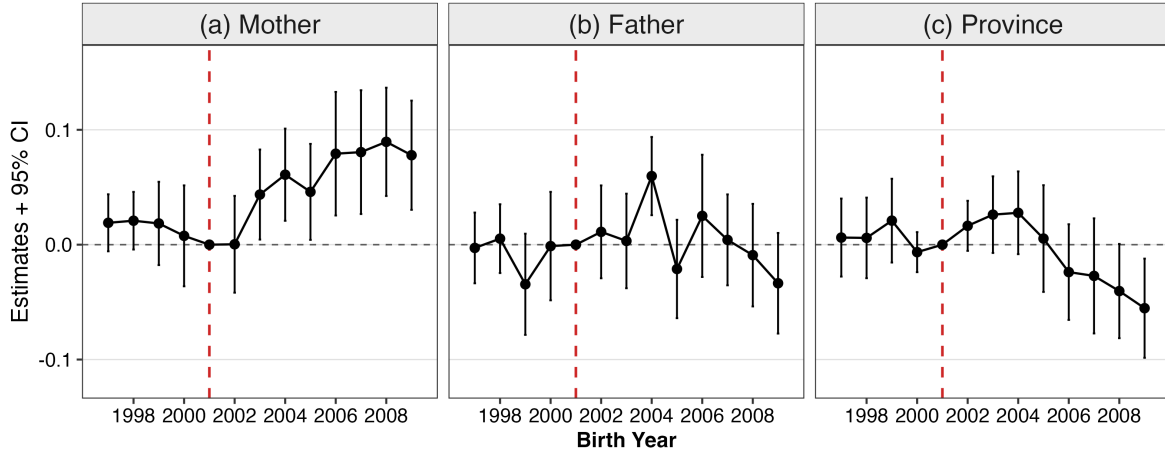
our magnitudes are relative comparisons between more and less exposed mothers, not the BTA's contribution to the national trend. We defer their aggregate interpretation to Section 7, because it turns on the fertility results reported below.

Figure 4 traces these effects over time, plotting event study estimates from the full specification where all three exposure channels enter simultaneously with the complete set of controls. Appendix Figure C2 presents individual event studies for each channel estimated separately, confirming that the patterns are not artifacts of joint estimation.

Panel (a) displays the maternal exposure coefficients. Pre-BTA estimates for the 1997–1999 Census cohorts and the 2000–2001 VHLSS births are individually and jointly indistinguishable from zero (joint test $p = 0.46$), confirming that mothers with different exposure intensities followed parallel trends before the shock. The 2002 coefficient is essentially zero. The maternal estimates turn positive and significant in 2003 and 2004, and remain significant in every year from 2005 through 2009. The timing is consistent with gestation. Because the BTA took effect in December 2001, most births recorded in 2002 reflect conceptions that predate the agreement, and affected births would appear in later cohorts.

Panels (b) and (c) display paternal and province-level exposure. Neither channel exhibits a consistent positive post-BTA pattern. Paternal industry exposure fluctuates around zero in most

Figure 4: Event Study Estimates of BTA Exposure on Male Birth Probability



Notes: Coefficients from Equation (13) with all three exposure channels and full controls. Panel (a) maternal, panel (b) paternal, panel (c) provincial exposure. Parent coefficients per 0.10 exposure increase, province per one standard deviation. Omitted year 2001 (vertical line). Bars are 95% confidence intervals, province-clustered.

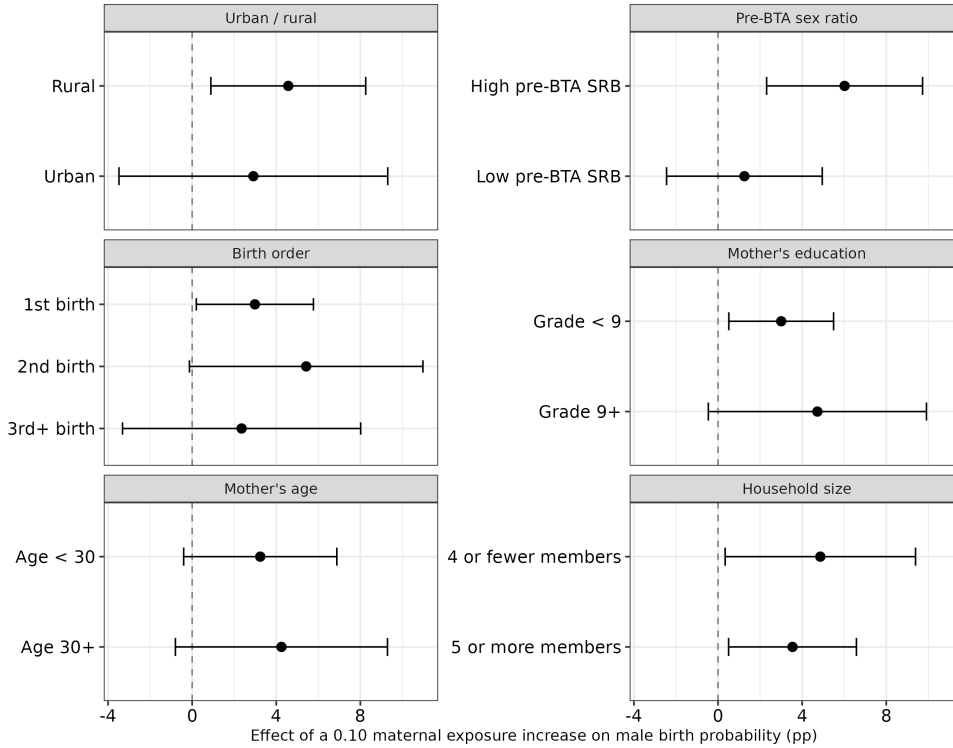
years, with the exception of an isolated 2004 spike.¹⁴ Province-level exposure drifts negative toward the end of the sample period. Pre-trend coefficients are individually insignificant for both channels, with joint tests of $p = 0.21$ for paternal and $p = 0.08$ for provincial exposure. The positive response is unique to maternal exposure and begins only when BTA-affected conceptions could reach term.

Heterogeneity. We examine how the maternal exposure effect varies across subgroups by estimating Equation (13) separately within each group, with the same fixed effects and controls as the main specification. Figure 5 displays the coefficient on $\tau_m \times \text{Post}_t$, scaled to a 0.10 exposure increase, and its 95 percent confidence interval for each subgroup.

The maternal exposure effect is positive in both rural and urban areas, with estimates of 0.046 and 0.029, respectively. The rural estimate is statistically significant ($p < 0.05$) while the urban estimate is imprecise. The significant rural response fits the norms reading of son preference. Patrilineal practice and traditional family obligations remain stronger in rural Vietnam, so the same increase in the maternal opportunity cost of childcare meets a stronger underlying demand for sons (Guilmoto, 2012). The pattern also does not fit a purely urban technology-access explanation, which would concentrate effects in cities where prenatal diagnosis is closest at hand. An alternative possibility is that rural households are poorer, in the sense that wage income is scarcer and a formal wage job changes the value of the mother's time by more.

¹⁴Annual event-study cells are small, so we do not interpret isolated single-year movements on their own. The 2004 paternal estimate is not followed by a persistent positive pattern in adjacent years, unlike the maternal series.

Figure 5: Heterogeneous Effects of Maternal Industry Exposure on Male Birth Probability



Notes: Each point is the maternal coefficient from Equation (13) within the indicated subgroup, per 0.10 exposure increase, with 95% province-clustered confidence intervals. Pre-BTA SRB is split at the median male share among 1997–1998 Census births. Household size for Census births is merged from IPUMS household records.

A more direct test of the norms reading uses revealed preference. We proxy preference intensity with the province's pre-BTA sex ratio, using the median male birth share among 1997–1998 Census births as the cutoff (Figure C3 maps the provincial variation). Provinces with higher pre-existing sex ratios show a larger maternal exposure effect (0.060, $p < 0.01$) than provinces with lower pre-BTA sex ratios (0.012, not significant). The split is another proxy for norms. Provinces differ persistently in the strength of patrilineal practice, strongest in the north, and the pre-BTA sex ratio summarizes where those norms already produced male-skewed births before the shock (Guilmoto, 2012).

We split by birth order because son preference concentrates selection pressure at particular parities, and the time-cost mechanism might locate that pressure differently. Selection is likely to concentrate at second and higher births in families whose earlier children are girls, because a family that already has a son faces little pressure at the next birth (Ebenstein, 2011; Anukriti et al., 2022). In our setting, with fertility near two births and a maternal time constraint that can bind from the first child, selection can appear at first parity. We find that the first-birth estimate is positive and significant (0.030, $p < 0.05$). The second-parity estimate is similar (0.054, $p < 0.10$),

the third-and-higher estimate is positive but imprecise (0.024), and the differences across parities are not statistically distinguishable.¹⁵ First-parity selection matches the aggregate evidence for Vietnam, where sex ratios are skewed already at first births (Guilmoto, 2012). The parity profile is informative about mechanisms. Selection driven by sibling composition predicts effects only after a firstborn girl, while the time-cost mechanism generates pressure from the first birth, which is what we find.

Mothers with higher education (ten or more years of schooling) show a larger point estimate (0.047, $p < 0.10$) than those with less schooling (0.030, $p < 0.05$). More educated mothers earn higher wages and are more likely to hold formal wage jobs, so the same industry-level improvement raises the value of their time by more and sharpens the incentive to economize on time-intensive childcare through sex selection. While the difference between the education groups is not statistically significant, the larger point estimate among more educated mothers is consistent with our model.

Maternal age at birth shows a modest gradient. Mothers aged 30 and above have a slightly larger point estimate (0.042) than mothers under 30 (0.032, $p < 0.10$). Older mothers are more likely to be at higher parities without yet having a son, have accumulated work experience that raises the opportunity cost of childcare, and face a shorter remaining fertility horizon. The pattern is directionally consistent with these forces, but mothers under 30 respond nearly as strongly, perhaps because most births occur to mothers in their twenties, when career entry and childbearing coincide and the time-cost pressure already binds.

The sixth split examines household size, which proxies for the availability of childcare substitutes within the home. The time-cost mechanism predicts stronger effects where fewer coresident adults can absorb childcare (Guo et al., 2018), and the point estimates lean that way. The estimate is 0.049 in households with four or fewer members ($p < 0.05$), the size of a nuclear household with two children, against 0.035 in households with five or more ($p < 0.05$). One caveat attaches to this split. Household size is measured at the survey and includes the children themselves, so it is partly an outcome of the fertility choices under study.

Overall, the heterogeneity results point to a stronger response when maternal wage opportunities interact with stronger incentives to secure a son. The first-birth estimate is positive, and the largest responses appear in rural areas and in provinces with high pre-BTA sex ratios.

5.2 Effects on Parental Labor Market Outcomes

We next examine directly the labor-market effects the BTA induced for the parents of the children studied above. Prediction 1 requires maternal exposure to raise the mother's labor-market returns, which we proxy with wage employment because the wage offer w_m is not observed, and Prediction 2 requires the mirror response for fathers. Table 4 estimates the same specification on

¹⁵Births of order 3 or higher are uncommon in our data. Vietnam's total fertility rate had already fallen to 2.01 by 2000.

Table 4: Effects of BTA Exposures on Parental Labor Market Outcomes

	MomWageEmp		DadWageEmp		log(MomWorkHrs)	
	(1)	(2)	(3)	(4)	(5)	(6)
$\tau_m \times \text{Post}$	0.063*** (0.015)	0.064*** (0.015)	-0.008 (0.015)	-0.007 (0.014)	0.030 (0.024)	0.033 (0.024)
$\tau_d \times \text{Post}$	-0.007 (0.007)	-0.007 (0.007)	0.033*** (0.010)	0.034*** (0.010)	0.015 (0.018)	0.016 (0.018)
$\tau_p \times \text{Post}$		-0.001 (0.023)		-0.024 (0.030)		-0.086** (0.040)
R^2	0.60	0.60	0.53	0.53	0.25	0.25
Observations	17,665	17,665	17,665	17,665	17,472	17,472
Dep. Var. Mean	0.28	0.28	0.44	0.44	4.68	4.68

Notes: Dependent variables are maternal wage employment (cols. 1–2), paternal wage employment (cols. 3–4), and log monthly maternal hours (cols. 5–6). Within each outcome, the second column adds province exposure. Entries are effects of a 0.10 increase in parent-specific exposure and a one-standard-deviation increase in province exposure. All columns include province, survey-year, and both parents' industry fixed effects, together with the full control set of Equation (13) matched to the survey year. Province-clustered standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

parental labor-market outcomes, with maternal wage employment in columns (1)–(2), paternal wage employment in columns (3)–(4), and maternal log monthly hours in columns (5)–(6).

Column (1) shows that a 0.10 increase in maternal exposure, roughly a 10 percentage-point larger tariff cut, raises the probability that the mother holds a wage job by 6.4 percentage points after the BTA ($p < 0.01$), about a quarter of the 28 percent sample mean. The estimate is unchanged in column (2) when provincial exposure is added, and paternal and province-level exposure effects on maternal wage employment are small and statistically insignificant.

Wage employment is the relevant labor-market measure in this setting because, although female labor force participation was approximately 70 percent before the BTA, only one in four mothers held a wage job (panel (b) of Figure 1), leaving most working mothers in informal or agricultural work without an observed market wage.¹⁶ The transition into wage employment from informal or agricultural work is itself a signal of improved labor market conditions, and the shift from the informal to the formal sector is precisely the reallocation that [McCaig and Pavcnik \(2018\)](#) document using the same tariff variation. In the Vietnamese context, formalization raises

¹⁶Wage regressions are infeasible in our data for two reasons. Wage income is observed only for wage earners, and wage-earner status is itself the treated margin, so a regression conditional on positive wage income mixes wage-offer changes with the changing composition of earners. Sample size compounds the problem. Only 862 women in the rotating panel report positive wage income on both sides of the BTA, 614 under the sex-selection sample's mother definition and 291 staying in the same industry, so minimum detectable wage effects are on the order of 40 percent.

the value of market work through more stable wage jobs while also making work schedules less flexible, both of which increase the opportunity cost of time spent away from employment.

On the intensive margin, columns (5)–(6) show a positive but imprecisely estimated maternal hours response of about 0.03 log points per 0.10 exposure increase ($p = 0.18$). Three features of the hours measure work against precision. The BTA shifts some mothers from informal or agricultural work into wage employment, so the hours estimate mixes changes within existing jobs with movement into jobs that carry more regular schedules. Monthly hours refer to the main job, so multiple job-holding adds further noise. Finally, every mother in this sample has an infant, so maternity leave and reduced schedules around the birth compress observed hours for reasons unrelated to exposure. The paternal coefficient on maternal hours is small and statistically indistinguishable from zero, close to the weakly negative response the model assigns to a pure income gain. The provincial coefficient for hours is negative and significant in column (6), and this sign is what Prediction 3 implies for maternal labor supply. Broad local exposure operates partly as an income gain, and an income gain buys leisure.

Columns (3)–(4) show the mirror pattern for fathers’ wage employment. Maternal industry exposure ($\tau_m \times \text{Post}_t$) is near zero for paternal wage employment, while a 0.10 increase in paternal industry exposure ($\tau_d \times \text{Post}_t$) raises the father’s wage-job probability by 3.4 percentage points ($p < 0.01$). The labor-market response is therefore parent-specific. Each parent’s wage employment responds to exposure in that parent’s own industry. This result verifies the labor-market first stage behind Prediction 2, but only the maternal channel transmits to sex selection.

5.3 Effects on Fertility

Predictions 1 and 2 generate opposing fertility effects. Maternal exposure reduces births through time costs, while paternal exposure raises births through income (Table 1, column 3). Fertility is a demanding outcome for our design. Over a short window, annual birth probabilities are small, while over a long horizon the requirement is heavier still, because each birth must be linked to the parents’ pre-birth industries and that link erodes with migration and industry switching. The reconstructed panel accepts the short-run tradeoff. Table 5 reports estimates from the reconstructed woman-year panel described in Section 4, where the dependent variable equals one if the woman gave birth in that year.

The retrospective woman-year panel follows Equation (15) with mother fixed effects. The estimates compare changes in a mother’s annual birth propensity around the BTA across exposure intensities. The maternal coefficient is negative and significant in columns (1) through (3) of Table 5, reaching -0.0012 per 0.10 exposure increase in the full specification ($p < 0.05$), about 7 percent of the sample mean annual birth probability of 1.8 percent. Because this outcome is reconstructed from children observed in the household roster, its mean should be read as the mean of an annual birth-hazard measure, not as an estimate of the total fertility rate. The sign is consistent with the model prediction that maternal industry exposure reduces fertility by raising the

Table 5: Within-Mother Evidence on Fertility and Sex Selection

	AnyBirth			MaleBirth (within mother)		
	(1)	(2)	(3)	(4)	(5)	(6)
$\tau_m \times \text{Post}$	-0.0008** (0.0004)	-0.0009*** (0.0003)	-0.0012** (0.0005)	0.097** (0.042)	0.109** (0.042)	0.110** (0.043)
$\tau_d \times \text{Post}$	0.002*** (0.0005)	0.002*** (0.0005)	0.003*** (0.0005)	0.077 (0.052)	0.080 (0.052)	0.079 (0.053)
$\tau_p \times \text{Post}$		0.0009*** (0.0003)	0.0038*** (0.0011)		-0.056 (0.051)	-0.052 (0.051)
Observations	2,011,963	2,011,963	2,011,963	2,615	2,615	2,615
Dep. Var. Mean	0.018	0.018	0.018	0.52	0.52	0.52
Identifying mothers				407	407	407
Mother FE	✓	✓	✓	✓	✓	✓
Birth Year FE	✓	✓	✓	✓	✓	✓
Controls			✓			✓

Notes: Columns (1)–(3) report Equation (15) on the reconstructed woman-year panel of mothers aged 20–40 from the VHLSS 2002–2008, where the dependent variable equals one if the woman gave birth in that year. Column (3) adds world-average tariff controls and pre-BTA sector-share-by-year interactions. Columns (4)–(6) re-estimate the male-birth specification on the VHLSS rotating panel with each parent’s exposure fixed at the 2002 wave, following the same build-up as columns (1)–(3). Identifying mothers are those observed with births both before and after the BTA. Entries are effects of a 0.10 increase in parent-specific exposure and a one-standard-deviation increase in province exposure. Province-clustered standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

opportunity cost of the mother’s time.

Paternal industry exposure moves in the opposite direction. Once maternal exposure is included, the paternal coefficient is 0.002 to 0.003 per 0.10 exposure increase ($p < 0.01$) across columns (1) through (3), between 11 and 17 percent of the mean annual birth probability. Both parental effects are modest against completed fertility. Cumulated mechanically over a 20-year reproductive window, a 0.10 exposure contrast implies on the order of 0.02 fewer births per woman through the mother and 0.05 more through the father, small shares of a total fertility rate near two. The positive sign is consistent with Prediction 2. Paternal exposure raises fertility through an income effect without tightening the maternal time constraint.

Province-level exposure ($\tau_p \times \text{Post}_i$) enters positively and significantly in columns (2) and (3), consistent with Prediction 3 when broader provincial exposure captures income gains outside the mother’s own job. Broad local gains raise births but leave the birth composition alone, exactly the division the model assigns to income.

The opposing signs of maternal and paternal exposure on fertility are difficult to reconcile with

a model where any positive income shock affects childbearing in the same direction. They instead point to the distinction between maternal time costs and household income. Maternal exposure raises the opportunity cost of childbearing and reduces births, while paternal exposure relaxes the household budget and raises births. The opposing signs also address the possibility that higher income could ease childbearing by allowing households to purchase childcare. If purchased childcare were a perfect substitute for the mother’s own time, a maternal wage gain would look more like a pure income shock and should raise fertility, as paternal exposure does. This pattern is consistent with childcare remaining a binding maternal time cost. Vietnam’s institutional context, in which formal childcare is scarce outside major urban centers and most working mothers rely on family or informal arrangements, reinforces this conclusion.

Two qualifications attach to these estimates. First, the outcome is an annual birth hazard, not completed fertility. Part of the estimated reduction may reflect delayed births rather than a permanent decrease in desired family size, and sex-selective abortion itself lengthens birth spacing, which would make the hazard effect larger than the completed-fertility effect. Second, the outcome is reconstructed from the household roster rather than from recorded birth histories. To gauge whether the reconstruction can be trusted, we benchmark it against the only Vietnamese surveys that record complete birth histories, the 1997 and 2002 DHS. The check asks whether the roster-based hazard reproduces the birth rates that mothers report directly. We compute the probability of a birth for each combination of mother’s age and calendar year in both datasets, so that a 25-year-old in 1998 in the roster is compared with a 25-year-old in 1998 in the DHS, and we average the cell differences using the roster’s weights. On these common cells, the roster hazard lies within 0.1 to 0.4 percentage points of the DHS hazard, on a base of 10 to 13 percent (Appendix Table C2). This result suggests that the reconstruction recovers the level of fertility where it can be verified. No DHS with birth histories exists for Vietnam after 2002, so the benchmark validates only pre-BTA levels.

5.4 Panel Evidence

The cross-sectional results align with the model, but they rely on parent industries measured at the time of the survey. The rotating panel addresses the specific sorting concern that women with stronger son preference moved into high-tariff-cut industries after the BTA, inflating the maternal exposure coefficient. The VHLSS re-interviews a subset of households across the 2002, 2004, and 2006 waves, so the same mother can be observed with births on both sides of the BTA. We re-estimate Equation (13) on this panel with two changes. Each parent’s exposure is fixed at the industry reported in the 2002 wave, so treatment predates any post-BTA switching, and mother fixed effects replace the province and industry fixed effects. The mother fixed effects absorb everything constant about a woman, including her baseline son preference and her permanent industry position, so the coefficient no longer compares different women. Because exposure is fixed and only its post-BTA interaction varies, what remains is the change within the same mother between

her pre-BTA and post-BTA births, which is why only mothers with births on both sides of the agreement contribute identifying variation. Fixing exposure at 2002 matters because switching was common. Among women working in 2002, approximately 23 percent changed their 2-digit ISIC industry by 2004 or 2006. Appendix Table C1 shows that the 2002 panel baseline is broadly similar to the full cross-section on education, ethnicity, urban residence, and provincial exposure, with panel mothers about one year older on average.

Columns (4) through (6) of Table 5 report the estimates, following the same build-up as the fertility columns. For reference, the cross-sectional maternal estimate on the same 2002–2006 sample with contemporaneous industry assignment is 0.022 per 0.10 exposure increase ($p < 0.05$, not shown). With mother fixed effects and 2002-fixed exposure, the maternal coefficient is 0.097 with maternal and paternal exposure in column (4) ($p < 0.05$), 0.109 adding provincial exposure in column (5) ($p < 0.05$), and 0.110 with demographic controls in column (6) ($p < 0.05$), while the paternal and provincial coefficients are indistinguishable from zero throughout.

Fixing 2002 exposure removes post-BTA industry switching as a source of the effect, and the maternal coefficient survives, so contemporaneous sorting does not generate the cross-sectional pattern. The within-mother magnitudes are, however, not our preferred estimates. They rest on the small set of mothers observed with births on both sides of the BTA, a selected group with unusually high fertility over a short window.

Taken together, the cross-sectional and panel results trace a discriminating pattern across the three exposure channels and three outcomes that maps directly onto the model predictions in Table 1. The evidence matches the full maternal row. Maternal industry exposure increases male birth probability, raises maternal wage employment, and reduces fertility, as Prediction 1 requires. The paternal results also fit the key sex-selection and fertility predictions. Paternal exposure does not move the male birth share but raises fertility, consistent with Prediction 2. Provincial exposure matches the predicted null for sex selection and shows a positive fertility response, consistent with Prediction 3 when local exposure operates partly as an income shock. The results support the model where it is most distinctive. The sex-selection response appears only when the shock reaches the mother’s own labor-market opportunity, and the two parents’ fertility responses move in opposite directions.

6. Robustness Checks

This section examines the sensitivity of our main results to alternative estimation choices. We consider alternative clustering schemes, region-by-year fixed effects, nonlinear specifications, and binary treatment definitions, together with dose diagnostics for the continuous specification, among other checks.

Alternative Clustering. Our baseline clusters standard errors by province (61 clusters). Although tariff cuts vary by industry, the household-level treatment combines maternal, paternal,

and provincial exposure. We also report clustering by 59 two-digit ISIC industries (Table C3), two-way clustering by province and industry (Table C4), and a restricted province-level wild-cluster bootstrap for the male-birth outcome (Appendix Table C5). The maternal male-birth coefficient of 0.037 per 0.10 exposure increase carries a standard error of 0.012 under both alternative clustering schemes, against 0.014 at baseline, with a wild-bootstrap p -value of 0.027. The maternal wage-employment estimate remains significant at the 1 percent level, the hours coefficient becomes marginally significant under industry clustering, and the provincial coefficients are stable. The one weakening is the maternal fertility coefficient, whose p -value rises above 0.10 under industry and two-way clustering, while the paternal fertility coefficient retains significance at the 1 percent level throughout.

Identifying Variation. Two related concerns are that a single industry drives the maternal estimate and that the effective number of exposure cells is small. Appendix Figure C4 drops one maternal industry at a time. Every estimate remains positive and inside the full-sample confidence interval, with food manufacturing the most influential omission (0.025 per 0.10 exposure increase, $p = 0.09$), so no single industry carries the result. The top five industries hold about four-fifths of the identifying mass (Appendix Table C5), which motivates the wild-bootstrap inference above.

Specification Build-Up. The maternal coefficient rises from 0.021 per 0.10 exposure increase with fixed effects only to 0.037 with the full control set. Appendix Tables C6 and C7 add the control blocks one at a time and decompose the movement following Gelbach (2016). The entire 0.016 gap comes from the world-average tariff controls, with demographics and sector-share trends contributing less than 0.001 each. Multilateral tariff movements were correlated with BTA exposure and worked against the BTA contrast, so omitting them masks part of the maternal effect. Adding potentially endogenous demographic controls, the mother's age and parity, changes nothing (Appendix Table C8).

Timing and Local Shocks. The remaining threat is a shock coinciding with the BTA, either a local development tracking exposed provinces or a later trade reform. Three checks on the male-birth outcome address it (Appendix Table C9). Province-by-birth-year fixed effects, which absorb every province-level shock, leave the maternal coefficient at 0.037 per 0.10 exposure increase ($p < 0.01$). Restricting to births before 2007 removes any overlap with WTO accession and gives 0.038, significant at the 5 percent level. Splitting the post period gives 0.034 for 2002–2005 and 0.064 for 2006–2009, a difference significant at the 5 percent level that mirrors the event-study path and gradual firm-level adjustment, while the early estimate shows the pre-WTO period alone carries the result.

Region-Year Fixed Effects. Regional infrastructure investment could independently affect access to prenatal technologies (Chen et al., 2013) or childcare, and such policies follow Vietnam’s eight administrative regions. Region-by-year fixed effects absorb this variation and, unlike the province-by-year check, can be applied to all four outcomes. Table C10 shows the estimates essentially unchanged, with the male-birth coefficient at 0.037 per 0.10 exposure increase ($p < 0.05$), wage employment at 0.061 ($p < 0.01$), and the opposing fertility responses retaining their signs and significance.

Nonlinear Estimation. Fixed-effects logit estimates reproduce the linear results (Table C11). The controlled maternal logit coefficients imply average marginal effects of 0.037 for the continuous exposure and 0.099 for the binary contrast, against linear estimates of 0.037 and 0.098, and the paternal and provincial coefficients remain small and insignificant, so the main estimates are not an artifact of the linear functional form.

Binary Treatment Definitions and Dose Diagnostics. Our main analysis uses the continuous tariff cuts. As Callaway et al. (2024) show, the two-way fixed effects coefficient on a continuous dose averages dose-specific effects with weights that can be negative below the mean dose, and a causal dose-response reading would require no selection on gains across doses, an assumption we do not impose. We therefore re-estimate the main contrasts with a binary high-versus-low definition, which identifies an average effect on the treated under standard parallel trends alone, reconcile the two scales under bootstrap inference, and check that the linear dose specification is conservative.

We classify each parent by the industry-level distribution of tariff cuts, which does not depend on how many workers each industry employs in a given data source. A parent is high-exposure when the industry’s tariff cut lies strictly above the median cut across industries. The high group consists mostly of manufacturing, and the low group combines lower-cut tradables with services, whose tariff cut is simply zero and whose workers participate in the same wage-labor market. Agricultural work instead enters as its own indicator for each parent, because its employment share changed sharply between the Census and the VHLSS waves and its labor market offers little wage employment, so pooling farmers into the low group would let composition changes redefine treatment. The provincial indicator splits at the provincial median. Appendix Table C15 reports the binary contrasts for all four outcomes in the same architecture as the continuous tables, and Appendix Figure C6 reports the corresponding event study. The results reproduce the continuous pattern. The maternal high-versus-low contrast is 0.098 for the male birth share ($p < 0.01$) and 0.076 for maternal wage employment ($p < 0.01$). The maternal contrast lowers the annual birth hazard (-0.0028 , $p < 0.05$) while the paternal contrast raises it (0.0042 , $p < 0.01$), and the paternal contrast raises paternal wage employment (0.047 , $p < 0.05$) without moving any maternal outcome. The event study shows a flat pre-period, with a joint test p -value of 0.52. Overall, the

binary and continuous designs agree on every sign.

Appendix Table C5 reconciles the two scales for the male-birth outcome. The binary contrast is 0.098 and the continuous estimate is 0.037 per 0.10 exposure increase, and both survive the restricted province-level wild-cluster bootstrap ($p = 0.015$ and $p = 0.027$). Finally, Appendix Figure C5 asks whether the linear-in-dose specification understates effects over the observed dose range of the continuous treatment. Quadratic, cubic, and piecewise specifications imply effects at a 0.10 dose at least as large as the linear estimate (0.076 to 0.088 against 0.037), so the linear specification is a conservative summary of the dose pattern. These curves are functional-form sensitivities, not causal dose-response estimates, which would require assumptions beyond standard parallel trends.

7. Discussion

The estimates so far trace a specific route. Maternal exposure raises maternal wage employment, lowers the annual birth hazard, and raises the male birth share, while paternal and provincial exposure raise fertility without moving sex selection. The model reads this joint pattern as the maternal time cost working through fertility. This section asks whether the alternative explanations, income, relative returns, bargaining, and postnatal composition, can account for the same pattern on their own, using the comparative statics in Table 1.

First, under the income-effects explanation (Almond et al., 2019), any income source should increase sex selection, including paternal wages and broad local gains captured by provincial exposure. Across all specifications, paternal and provincial exposure coefficients for sex selection are small and statistically insignificant (Table 3), even though these channels show the expected effects on other margins such as wage employment and fertility. The cross-channel pattern is difficult to reconcile with a model in which income alone drives sex selection. The same separation speaks to the designs that identify from household-level or cohort-level variation, which cannot assign the shock to a specific parent. Our evidence locates the prenatal response in the mother's own labor-market position, a margin those designs bundle. A variant of the income explanation holds that maternal earnings relax an affordability or access constraint on prenatal diagnosis and abortion. As discussed in Section 3.1, diagnosis and abortion were inexpensive and widely available through the public system before the BTA, and an affordability constraint would likely bind equally for paternal or non-labor income, which show no sex-selection response.

The relative-returns mechanism (Qian, 2008; Rosenzweig and Schultz, 1982) predicts that expanding opportunities in female-intensive industries should reduce sex selection by making daughters more valuable. Our positive maternal coefficient does not contradict this mechanism. It shows that in our setting the time-cost channel outweighs the value channel. The value channel is present in our model through the quantity-quality margin. Higher returns to daughters raise desired investment per child, which raises the cost of each birth and reinforces the fertility-pressure logic of Jayachandran (2017). In this sense our estimates build on the relative-returns insight. What they

reject is the net prediction that rising female returns reduce sex selection.

The preceding test interprets the relative-returns channel at the level of the mother’s own wage, but the mechanism’s original argument operates at the level of the local labor market. When female-intensive industries expand in a province, all families there may expect daughters to face better adult prospects regardless of the mother’s own industry, and Appendix A.2.1 generalizes the model to allow province-level returns to daughters to rise independently of the maternal wage. To test this local-market reading, we add to Equation (13) a female-weighted provincial exposure, $\tau_p^w = \sum_j f_j s_{jp} \tau_j$ with f_j the pre-BTA female employment share of industry j , standardized to standard-deviation units. The relative-returns prediction is a negative coefficient for sex selection, and the data do not deliver it. In Table C12, the female-weighted coefficient is statistically indistinguishable from zero for the male birth share, and the maternal coefficient is unchanged at 0.037 per 0.10 exposure increase when the fourth channel is added. The local-market version of the relative-returns mechanism therefore does not account for the male-birth response, which remains tied to the mother’s own industry exposure.

Another potential explanation is bargaining. Unlike income or relative returns, bargaining does not make a sharp prediction from the direction of the wage shock alone because the effect depends on whose preferences receive more weight. A rise in maternal wages could increase mothers’ bargaining power, and the resulting shift in household decision weights could affect fertility or sex selection even if childcare time costs did not change. To explain our main result, however, the bargaining channel would need stronger maternal bargaining power to raise son selection while also reducing fertility. That logic has two empirical implications. First, paternal wage gains should move bargaining power in the opposite direction and therefore should move sex selection, at least if the paternal shock is strong enough to affect household decisions. Second, households where mothers already have stronger bargaining positions should display more son selection before the trade shock.

The data do not support either implication. Paternal BTA exposure produces meaningful economic effects. A 0.10 increase in paternal exposure raises the father’s wage-job probability by 3.4 percentage points (Table 4) and raises the annual birth probability (Tables 5 and C12). Yet the paternal exposure coefficient for sex selection remains statistically indistinguishable from zero across the continuous (Table 3), binary (Appendix Table C15), and panel (Table 5) specifications. We also test whether pre-BTA bargaining position predicts sex selection using the 1999 Census. Among births in 1997 and 1998, mothers with strictly more education than their husbands, our coarse proxy for stronger maternal bargaining position, are no more likely to have a son (Table C13). These tests make bargaining unlikely to be the main driver of the pattern we observe.

A final alternative is postnatal composition rather than prenatal sex selection. We observe children aged 0–1 who are alive and co-resident at the time of the survey, so maternal exposure could in principle change the observed male share through sex-differential infant mortality, non-co-residence, or selective fostering. The mortality channel would require female infant mortality

to rise relative to male infant mortality. Aggregate infant-mortality trends point in the opposite direction. Between 2002 and 2008, male and female infant mortality declined nearly symmetrically, and male mortality remained higher throughout (Appendix Table C14). Fostering of infants aged 0–1 is also rare in Vietnamese household structure. Administrative birth records linking ultrasound visits to maternal employment would provide a definitive test, but the available evidence suggests that postnatal composition is more likely a measurement limitation than the main explanation for the cross-channel pattern.

A last question concerns what the magnitudes mean in the aggregate. Our estimates are relative comparisons between more and less exposed mothers under the same agreement, and they likely overstate the BTA’s contribution to Vietnam’s national trend, for two reasons the estimates themselves supply. The mechanism runs through fertility, yet in the aggregate the BTA loosened rather than tightened the fertility bottleneck. Scaled by mean exposure, the paternal fertility effect is roughly double the maternal one in the opposite direction, so the average household faced weaker fertility pressure than the maternal contrast alone implies. The value channel points the same way. If expectations about daughters’ adult earnings form from the local labor market rather than from the mother’s own wage, the higher returns to daughters that the export boom created reached exposed and unexposed mothers alike, were differenced out of the estimates, and pushed aggregate selection down. A force could also run the other way. If wage gains in exposed industries spread to other employers competing for the same workers, then the comparison group’s wages rose too, the gap between exposed and unexposed mothers understates the effect of the underlying wage gain, and the aggregate could exceed what the relative contrast suggests. Our design cannot net these opposing forces. Mothers with little measured exposure were also reached through their husbands’ industries and provincial demand, and the BTA was one episode in a broader liberalization, alongside the 2000 Enterprise Law and expanding foreign investment, that moved the same maternal margin.

8. Concluding Remarks

This paper shows that better jobs for mothers can increase sex selection at birth when son preference is present and childcare remains primarily maternal. Higher maternal wages raise the opportunity cost of childbearing, reduce fertility, and increase pressure to concentrate births on sons. The quantity-quality response helps reconcile this pattern with rising investment in girls who are born. Greater daughter returns can increase desired human-capital investment even as the sex ratio at birth becomes more male. Empirically, maternal exposure to the 2001 US-Vietnam Bilateral Trade Agreement raises male birth probability and modestly lowers the annual birth hazard. Paternal exposure raises paternal wage employment and fertility without moving sex selection. Provincial exposure does not move sex selection either.

The mechanism we identify is not unique to Vietnam. Albania, Montenegro, and Georgia show similar trajectories around their own trade integration episodes (Figure 1), and the same

structural conditions are present in the belt of countries where son preference, patrilocal family structures, and prenatal sex-selection technology coincide ([Guilmoto, 2012](#)). In Vietnam, the effects concentrate in rural areas and in provinces with higher pre-existing sex ratios, suggesting that the mechanism is most operative where son preference is strongest. Because the mechanism operates through the market opportunities of mothers already attached to the labor force, it speaks less directly to settings with low female participation, such as northern India. More broadly, where mothers bear the primary childcare burden and cultural or institutional norms favor sons, new female employment opportunities can narrow economic gender gaps while widening demographic gender imbalance. General-equilibrium forces are likely to dampen the mechanism over longer horizons. Rising maternal wages call forth childcare substitutes, a scarcity of brides raises the value of daughters through the marriage market, and daughters who earn and remit factory wages erode the old-age-support rationale for sons ([Guo and Zhang, 2020](#)), so the relationship we estimate over a short window need not persist at the same strength.

Addressing the link between female employment and sex selection requires policies that reduce the maternal time constraint directly. Income transfers or wage subsidies alone will not suffice, since the mechanism runs through the opportunity cost of mothers' time rather than household income. The human-capital extension shows that education costs can amplify the mechanism, but education-price policy is not a clean substitute for childcare because it can also change the relative cost of selecting sons. Future work with administrative birth records and direct measures of childcare access could test more precisely whether institutional childcare weakens the link between maternal employment opportunities and sex selection.

Generative AI Disclosure Statement

During the preparation of this work, the authors used OpenAI’s GPT-4 and GPT-5, Grammarly, Refine.Ink, and Anthropic’s Claude models to assist with producing computer code for data cleaning and analysis, identifying relevant research articles, outlining the structure of the paper, drafting candidate sentences for inclusion, and improving the organization, conciseness, and clarity of the writing. All AI-generated content was reviewed and edited by the authors, who take full responsibility for the final article.

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Online Appendix to Better Jobs for Mothers, More Sons at Birth

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A. Model Appendix

This appendix provides the full derivations supporting the model in Section 2. Appendix A.1 derives the first-order conditions and comparative statics for the baseline model via Slutsky decomposition. Appendix A.2 extends the model to include human capital investment in a quantity-quality framework.

A.1 Baseline Model Derivations

A.1.1 Lagrangian and First-Order Conditions

The household maximizes $U(c, \ell, n, B)$ with $B = bn$, choosing (c, ℓ, n, b) subject to the budget constraint $c + w_m \ell + w_m \gamma n + p_b b n = W$. The Lagrangian is

$$\mathcal{L} = U(c, \ell, n, bn) + \lambda [W - c - w_m \ell - w_m \gamma n - p_b b n]. \quad (\text{A1})$$

The first-order conditions with respect to c , ℓ , n , and b are

$$U_c = \lambda, \quad (\text{A2})$$

$$U_\ell = \lambda w_m, \quad (\text{A3})$$

$$U_n + U_B b = \lambda (w_m \gamma + p_b b), \quad (\text{A4})$$

$$U_B = \lambda p_b. \quad (\text{A5})$$

The fertility condition (A4) includes $U_B b$ on the left (the marginal benefit of one more child through the boy channel, since each child yields b expected boys) and $p_b b$ on the right (the sex-selection cost per child). Substituting Equation (A5) into Equation (A4) yields

$$U_n = \lambda w_m \gamma. \quad (\text{A6})$$

At the optimum, the effective shadow price of fertility is $w_m \gamma$, the maternal time cost alone. The household simultaneously optimizes b , so the sex-selection cost of an additional child ($p_b b$) is exactly offset by the marginal boy value ($U_B b$). Dividing through by $U_c = \lambda$, the MRS conditions reported in the main text (Equation 4) follow.

The effective price structure $(1, w_m, w_m \gamma, p_b)$ for goods (c, ℓ, n, B) permits standard Slutsky decomposition.

Maternal wage effect on sex selection. The maternal wage w_m enters two prices, $p_\ell = w_m$ and $p_n = w_m\gamma$. Thus, the compensated effect on the sex ratio $b = B/n$ operates through both channels:

$$\frac{\partial b^S}{\partial w_m} = \underbrace{1 \cdot \frac{\partial b^S}{\partial p_\ell}}_{\text{leisure-price channel}} + \underbrace{\gamma \cdot \frac{\partial b^S}{\partial p_n}}_{\text{fertility-price channel}} > 0, \quad (\text{A7})$$

where positivity holds when the boy share b and fertility n are Hicksian substitutes ($\partial b^S/\partial p_n > 0$). The household raises the boy share rather than just cutting the number of children when children become more expensive. The same logic appears in the literature (Jayachandran, 2017): with son preference, smaller families concentrate desired sons into a higher share. The total (Marshallian) effect adds the income effect:

$$\frac{\partial b}{\partial w_m} = \underbrace{\frac{\partial b^S}{\partial w_m}}_{> 0 \text{ (substitution)}} + \underbrace{h_m \frac{\partial b}{\partial W}}_{\text{income effect on } b}, \quad (\text{A8})$$

where $h_m = 1 - \ell - \gamma n$ is the mother's labor supply. The income effect on b is equivalent to $h_m \left(\frac{1}{n} \frac{\partial B}{\partial W} - \frac{b}{n} \frac{\partial n}{\partial W} \right)$ via the chain rule on $b = B/n$. In elasticity terms, $\eta_b^W = \eta_B^W - \eta_n^W$: the income effect on b is positive when $\eta_B^W > \eta_n^W$ (boys more income-elastic than fertility), zero when they coincide, and negative otherwise. Under son preference, it is likely that marginal dollars tilt toward the boy share, so $\eta_B^W \geq \eta_n^W$. With substitution positive and income weakly positive, $\partial b/\partial w_m > 0$.

Maternal wage effect on fertility. Both price channels reduce fertility:

$$\frac{\partial n}{\partial w_m} = \underbrace{\frac{\partial n^S}{\partial w_m}}_{< 0 \text{ (substitution)}} + \underbrace{h_m \frac{\partial n}{\partial W}}_{\text{income effect}}. \quad (\text{A9})$$

The substitution effect is unambiguously negative: own-price Slutsky $\partial n^S/\partial p_n \leq 0$ enters the chain rule with coefficient γ . The income effect $h_m \cdot \partial n/\partial W$ may partially offset if n is a normal good, but its force is scaled by h_m , the mother's net labor supply. When γ is significant because childcare is materially time-intensive, the substitution term dominates and $\partial n/\partial w_m < 0$.

Maternal wage effect on labor supply. The time constraint $h_m = 1 - \ell - \gamma n$ pins down labor supply as a residual. Substituting the Slutsky decompositions of $\partial \ell/\partial w_m$ and $\partial n/\partial w_m$:

$$\frac{\partial h_m}{\partial w_m} = - \underbrace{\left[\frac{\partial \ell^S}{\partial w_m} + \gamma \frac{\partial n^S}{\partial w_m} \right]}_{\text{substitution: } > 0} - \underbrace{h_m \left[\frac{\partial \ell}{\partial W} + \gamma \frac{\partial n}{\partial W} \right]}_{\text{income: } < 0 \text{ if } \ell, n \text{ normal}}. \quad (\text{A10})$$

This decomposition exhibits the classic backward-bending labor supply tension where substitution and income work in opposite directions. The same condition that signs $\partial n / \partial w_m$, i.e. substitution dominates because γ amplifies the time-price terms, signs labor supply: $\partial h_m / \partial w_m > 0$.

Paternal wage effect. The paternal wage w_d enters no price. Its effect on any demand operates entirely through full income W :

$$\frac{\partial b}{\partial w_d} = \frac{\partial b}{\partial W}. \quad (\text{A11})$$

By the same elasticity decomposition $\eta_b^W = \eta_B^W - \eta_n^W$ used for the maternal income effect, $\partial b / \partial W$ is positive when boys are more income-elastic than fertility, zero when the elasticities coincide, and negative otherwise. Under son preference, the positive case is plausible because marginal dollars may tilt toward birth composition. The critical asymmetry does not require this income effect to be large. The maternal wage adds substitution effects on top of the income effect, while the paternal wage has income only.

The paternal wage, via income effect, also raises fertility ($\partial n / \partial w_d > 0$ when n is normal) and reduces maternal hours ($\partial h_m / \partial w_d < 0$, since leisure is normal). Non-labor income I has the identical structure where $\partial b / \partial I = \partial b / \partial W$ and likewise for the other outcomes.

A.2 Human Capital Extension

We now extend the baseline model to let parents also choose how much to invest in each child, separately for sons and daughters. The choice set expands from (c, ℓ, n, b) to $(c, \ell, n, b, e_b, e_g)$, where e_b and e_g are the educational inputs (e.g., years of schooling) per son and per daughter, purchased at unit price p_e . Each unit of input is converted into human capital through a production function $h(\cdot)$ with $h' > 0$, $h'' < 0$, and $h(0) = 0$. The return to a unit of human capital is r_b for sons and r_g for daughters; in the baseline mapping, $r_b = w_d$ and $r_g = w_m$. Average per-child human capital is $\bar{E} = b r_b h(e_b) + (1 - b) r_g h(e_g)$, and parents value it through a separable, increasing, concave function $V(\bar{E})$.

The household problem is to maximize $U(c, \ell, n, B) + V(\bar{E})$, with $B = bn$, subject to the budget constraint

$$c + w_m \ell + n[w_m \gamma + p_e(b e_b + (1 - b) e_g)] + p_b b n = W. \quad (\text{A12})$$

Relative to the baseline budget, education spending $p_e[b e_b + (1 - b) e_g]$ enters per child, scaling with n . The Lagrangian is

$$\mathcal{L} = U(c, \ell, n, B) + V(\bar{E}) - \lambda [c + w_m \ell + n(w_m \gamma + p_e(b e_b + (1 - b) e_g)) + p_b b n - W]. \quad (\text{A13})$$

The first-order conditions for education investment are

$$\frac{\partial \mathcal{L}}{\partial e_b} = V'(\bar{E}) \cdot b r_b h'(e_b) - \lambda n b p_e = 0, \quad (\text{A14})$$

$$\frac{\partial \mathcal{L}}{\partial e_g} = V'(\bar{E}) \cdot (1 - b) r_g h'(e_g) - \lambda n (1 - b) p_e = 0. \quad (\text{A15})$$

Dividing by b and $(1 - b)$ respectively and equating yields the education condition stated in the main text:

$$r_b h'(e_b^*) = r_g h'(e_g^*). \quad (\text{A16})$$

Because $h'' < 0$, the gender with higher returns receives more investment. When men earn more than women in the labor market ($w_d > w_m$), the marginal return to a son's human capital exceeds the marginal return to a daughter's, so optimal investment per son exceeds optimal investment per daughter ($e_b^* > e_g^*$). Before 2005, the female-to-male ratio of university enrollment lay well below one (Figure 1, panel d), consistent with parents allocating more education to sons than to daughters under the prevailing gender wage gap.

The first-order condition for b is

$$U_B \cdot n + V'(\bar{E}) \cdot [r_b h(e_b) - r_g h(e_g)] = \lambda n [p_b + p_e(e_b - e_g)]. \quad (\text{A17})$$

The left-hand side is the marginal benefit of raising b . The first term ($U_B \cdot n$) is the direct preference for sons. The second term captures how shifting the sex composition toward boys changes average human capital. When sons are more valuable than daughters ($r_b h(e_b) > r_g h(e_g)$), this term is positive, adding to the incentive for sex selection. The right-hand side is the marginal cost: the sex-selection technology cost p_b plus the education premium $p_e(e_b - e_g)$ for each son over each daughter, summed over n children.

The first-order condition for n is

$$U_n + U_B b = \lambda [w_m \gamma + p_e(b e_b + (1 - b) e_g) + p_b b]. \quad (\text{A18})$$

Relative to the baseline fertility condition (Equation A4), the right-hand side now includes the per-child education cost $p_e(b e_b + (1 - b) e_g)$. An additional child requires not only the time cost $w_m \gamma$ and the sex-selection cost $p_b b$ but also investment in the new child's human capital, weighted by the expected sex composition. This term is the channel through which education prices feed back into the cost of fertility.

From Equation (A14), dividing by b :

$$V'(\bar{E}) \cdot r_b h'(e_b) = \lambda n p_e. \quad (\text{A19})$$

From Equation (A15), dividing by $(1 - b)$:

$$V'(\bar{E}) \cdot r_g h'(e_g) = \lambda n p_e. \quad (\text{A20})$$

These expressions show that sex composition does not enter the education FOCs directly once

we scale by the relevant child share. It enters indirectly through the shadow value λ . When $r_b > r_g$, optimal investment satisfies $e_b^* > e_g^*$. Holding (n, e_b^*, e_g^*) fixed, the education bill is $C_E = np_e[be_b^* + (1-b)e_g^*]$, so

$$\frac{\partial C_E}{\partial b} = np_e(e_b^* - e_g^*) > 0.$$

Replacing a daughter with a son raises the education bill when $e_b^* > e_g^*$, so a son carries a quality premium beyond the sex-selection cost p_b . Symmetrically, higher investment per child raises the marginal cost of an additional child through (A18). This two-way dependence is the quantity-quality tradeoff of [Becker and Lewis \(1973\)](#), and it is the channel through which the education margin enters both the fertility price and the sex-selection price below.

Rewriting the budget as in Equation (10), the effective prices are

$$p_c = 1, \quad p_\ell = w_m, \quad p_n = w_m\gamma + p_e e_g, \quad p_B = p_b + p_e(e_b - e_g). \quad (\text{A21})$$

Relative to the baseline (where $p_n = w_m\gamma$ and $p_B = p_b$), the maternal wage now enters p_n through two channels and the paternal wage enters p_B through the son education cost differential. The comparative statics follow the Slutsky decomposition of [Appendix A.1](#), applied to these modified prices.

Maternal wage effect on sex selection. A rise in w_m raises $r_g = w_m$. Holding the shadow cost of quality fixed, the direct return effect raises daughter investment because $h'' < 0$, narrowing the education gap. Indirect feedback through the shadow value of income λ and fertility n also moves both e_b^* and e_g^* . Provided the direct return effect dominates this feedback, the effect of w_m on prices is

$$\frac{dp_n}{dw_m} = \gamma + p_e \frac{de_g^*}{dw_m} > \gamma, \quad (\text{A22})$$

$$\frac{dp_B}{dw_m} = p_e \left(\frac{de_b^*}{dw_m} - \frac{de_g^*}{dw_m} \right). \quad (\text{A23})$$

The fertility price p_n then rises by a larger magnitude than in the baseline (where $dp_n/dw_m = \gamma$), because daughter education costs increase alongside time costs. The sex-selection price p_B declines when the education gap $e_b^* - e_g^*$ narrows.

In the baseline model, w_m enters only through prices in the budget constraint, and the Slutsky decomposition fully captures the substitution effect. With the human capital extension, w_m also enters the utility function through $r_g = w_m$ in $V(\bar{E})$. The maternal wage therefore affects b through two distinct routes. The first route operates through the budget constraint: w_m changes the effective prices, and the household reoptimizes along the budget. The price-side effect paral-

lels Equation (A7) with the modified prices:

$$\left. \frac{\partial b^S}{\partial w_m} \right|_{\text{HC}} = \underbrace{1 \cdot \frac{\partial b^S}{\partial p_\ell}}_{\text{leisure price}} + \underbrace{\frac{dp_n}{dw_m} \cdot \frac{\partial b^S}{\partial p_n}}_{\text{fertility price (amplified)}} + \underbrace{\frac{dp_B}{dw_m} \cdot \frac{\partial b^S}{\partial p_B}}_{\text{sex-selection price (new)}}. \quad (\text{A24})$$

In the baseline model, only the first two terms appear. The human capital extension adds the education component to the fertility-price channel and introduces the third term. We call this entire price-side mechanism the *cost channel*. Rising w_m makes children more expensive to raise and sons relatively cheaper, pushing b up. The first two terms of the cost channel are positive under the baseline condition that fertility and boys are Hicksian substitutes ($\partial b^S / \partial p_n > 0$). The third term is positive when the compensated own-price effect on boys is negative ($\partial b^S / \partial p_B < 0$) and the education gap narrows ($dp_B / dw_m < 0$). When both conditions hold, all three cost-channel forces reinforce.

The second route operates through the utility function, not the budget. Because $r_g = w_m$, the maternal wage also enters $V(\bar{E})$ directly. From the b -FOC (Equation A17), the marginal benefit of sex selection includes $V'(\bar{E}) \cdot [r_b h(e_b) - r_g h(e_g)]$. When w_m rises, r_g increases, so daughters' human capital $r_g h(e_g^*)$ rises and the gap $r_b h(e_b) - r_g h(e_g)$ narrows. The marginal benefit of selecting for sons falls. We call this the *value channel*. When daughters' future earnings prospects improve, each daughter becomes more valuable, and the household has less reason to skew births toward sons.

The total effect of w_m on b combines both routes:

$$\frac{db}{dw_m} = \underbrace{\left. \frac{\partial b^S}{\partial w_m} \right|_{\text{HC}}}_{\substack{\text{cost channel} \\ (+, \text{ through prices})}} + \underbrace{\left. \frac{\partial b}{\partial r_g} \right|_{\text{prices fixed}}}_{\substack{\text{value channel} \\ (-, \text{ through } V)}} + \text{income effect}. \quad (\text{A25})$$

These forces deliver the positive maternal prediction in Table 1. Higher maternal wages raise the time price of children, raise the education component of the fertility price, and lower the relative education-cost penalty of selecting sons when the gender investment gap narrows. The daughter-value channel works in the opposite direction, but it is unlikely to overturn these price-side forces to make the maternal effect zero or negative. The model therefore predicts $\partial b / \partial w_m > 0$ in settings where childcare remains materially maternal and fertility can adjust.

Maternal wage effect on fertility. In the baseline, $dn / dw_m < 0$ when the substitution effect dominates the income effect. The human capital extension strengthens this substitution force when daughter investment rises with the maternal wage, because the fertility price rises from $p_n = w_m \gamma$ to $p_n = w_m \gamma + p_e e_g$.

Paternal wage effect. As with the maternal wage, w_d affects b through both prices and utility. On the price side, w_d enters p_B directly through sons' returns and can enter p_n indirectly through the shadow-cost feedback:

$$\frac{dp_B}{dw_d} = p_e \left(\frac{de_b^*}{dw_d} - \frac{de_g^*}{dw_d} \right), \quad \frac{dp_n}{dw_d} = p_e \frac{de_g^*}{dw_d}. \quad (\text{A26})$$

Higher w_d raises sons' returns ($r_b = w_d$), so it directly raises e_b^* and widens the gap $e_b - e_g$, raising the sex-selection price p_B . The fertility price p_n moves only through indirect feedback on e_g^* and is therefore second-order. The price-side channel thus has a primary negative force through p_B and a possible offsetting positive force through p_n :

$$\left. \frac{\partial b^S}{\partial w_d} \right|_{\text{HC}} = \underbrace{\frac{dp_B}{dw_d} \cdot \frac{\partial b^S}{\partial p_B}}_{\text{sons costlier}} + \underbrace{\frac{dp_n}{dw_d} \cdot \frac{\partial b^S}{\partial p_n}}_{\text{quality-price feedback}}. \quad (\text{A27})$$

The first term is negative when sons become more expensive to educate. The second term, operating through the indirect quality-price channel, is positive when p_n rises. On the utility side, w_d enters $V(\bar{E})$ through $r_b = w_d$. When w_d rises, sons' human capital $r_b h(e_b^*)$ increases, widening the gap $r_b h(e_b) - r_g h(e_g)$ in the b -FOC (Equation A17). The marginal benefit of selecting for sons rises. This value channel is positive, the mirror image of the maternal case. The income effect is also weakly positive (Equation A11). The total effect is therefore:

$$\frac{db}{dw_d} = \underbrace{\left. \frac{\partial b^S}{\partial w_d} \right|_{\text{HC}}}_{\substack{\text{cost channel} \\ (-)}} + \underbrace{\left. \frac{\partial b}{\partial r_b} \right|_{\text{prices fixed}}}_{\substack{\text{value channel} \\ (+)}} + \text{income effect (+)}. \quad (\text{A28})$$

The human-capital extension therefore makes the paternal prediction weaker than the baseline pure-income prediction. Higher paternal wages raise sons' returns, which increases the value of sons but also raises the education-cost premium of selecting sons. The value channel and the income effect push b up, while the son-cost channel pushes b down. Because higher paternal wages do not raise the maternal time price of children, they lack the main force that drives the maternal prediction. The model therefore predicts a small or null paternal effect on sex selection.

The fertility prediction $\partial n / \partial w_d > 0$ is likely positive. The direct income effect raises fertility, as in the baseline. The human capital extension can introduce a partially offsetting force if higher w_d raises p_n through the indirect quality-price channel (Equation A26), creating a substitution penalty on quantity. Because the direct positive income effect is the leading force and the indirect quality-price feedback is second-order, the net effect on fertility is likely positive.

Non-labor income effect. Non-labor income I enters only the budget constraint. The cost and value channels that operate through parental wages are absent for I , leaving only the income

effect: $\partial b/\partial I = \partial b/\partial W$ and $\partial n/\partial I = \partial n/\partial W$. Under the income-elasticity condition above, the effect on b is weakly positive. Without any price-side force, the model predicts a small effect on sex selection and a positive effect on fertility.

A.2.1 When Returns Differ from Individual Wages

The main text sets $r_g = w_m$ and $r_b = w_d$, identifying children's returns with same-sex parental wages. This subsection relaxes that restriction. When female-intensive industries expand in a province, the local return to daughters' human capital may rise for all families, regardless of individual mothers' industries. Let r_g now denote this province-level return, which may differ from any individual mother's wage w_m .

The education FOCs carry through with r_g and r_b as separate parameters. The education condition becomes $r_b h'(e_b^*) = r_g h'(e_g^*)$, and the quantity-quality interaction (Equation A20) still determines optimal investment, so e_g^* depends on both r_g and n . The modified prices become $p_n = w_m \gamma + p_e e_g(r_g, n)$ and $p_B = p_b + p_e(e_b(r_b, n) - e_g(r_g, n))$. The key difference from the individual-level case is that w_m still determines the time cost of childcare ($w_m \gamma$ in p_n), but r_g now independently determines the education investment in daughters.

When r_g rises independently of w_m , two opposing forces emerge. The cost channel raises sex selection: higher r_g increases optimal investment e_g^* , raising the fertility price p_n and lowering the sex-selection price p_B . The value channel lowers sex selection: daughters' economic value rises in $\bar{E}$, reducing the household's willingness to select for sons (through the b -FOC, Equation A17). Unlike the maternal wage case, there is no time-cost force ($w_m \gamma$) giving the cost channel a built-in advantage, because r_g moves independently of w_m . The net effect is therefore more ambiguous:

$$\frac{db}{dr_g} = \underbrace{\frac{\partial b}{\partial r_g} \Big|_{\text{cost}}}_{>0} + \underbrace{\frac{\partial b}{\partial r_g} \Big|_{\text{value}}}_{<0}. \quad (\text{A29})$$

This decomposition motivates the empirical test using female-weighted provincial exposure P_p^w in Section 7. The measure P_p^w captures province-level variation in returns to female labor that operates independently of individual maternal wages. If the cost and value channels approximately offset each other, P_p^w can improve female labor-market opportunities, and potentially raise desired investment in daughters through e_g^* , without producing a detectable net effect on sex selection. The maternal wage effect can still be positive because, when $r_g = w_m$, the shock also raises the maternal time cost of children; P_p^w shifts daughters' local returns without directly tightening that maternal time constraint.

A.2.2 Binding Fertility Constraints

The baseline model treats fertility as a choice variable. When policy constrains fertility, as under China's one-child policy, the fertility margin is restricted and the model's predictions change. Suppose n is fixed at $\bar{n}$. The household chooses (c, ℓ, b, e_b, e_g) to maximize $U(c, \ell, \bar{n}, B) + V(\bar{E})$ subject to

$$c + w_m \ell + \bar{n} [w_m \gamma + p_e (b e_b + (1 - b) e_g)] + p_b b \bar{n} = W. \quad (\text{A30})$$

The education FOCs are unchanged. The education ratio (Equation A16) and the quantity-quality interaction (Equation A20) both carry through with $\bar{n}$ replacing n . The critical difference is that $\bar{n}$ is a parameter, not a choice variable. From Equation (A20), the direct shadow cost of quality is $\lambda \bar{n} p_e$. Holding the shadow value of income fixed, a lower $\bar{n}$ reduces this direct cost and supports higher investment per child. When the constraint binds at a low level (such as one child), this quantity-quality force pushes toward heavier investment in each child. If sons have higher returns ($r_b > r_g$), the education FOC implies $e_b^* > e_g^*$ and raises the education-cost premium for selecting a son, $p_e(e_b^* - e_g^*)$.

The constrained model generates three predictions that differ from the unconstrained case. First, constrained fertility raises the level incentive for sex selection but changes the composition of the maternal wage effect. The b -FOC under fixed fertility is

$$U_B + \frac{V'(\bar{E})}{\bar{n}} [r_b h(e_b) - r_g h(e_g)] = \lambda [p_b + p_e (e_b - e_g)]. \quad (\text{A31})$$

A low $\bar{n}$ raises the marginal value of an additional boy when preferences over boys are concave, because $B = b\bar{n}$ is small and $U_B(B)$ is high. It also scales up the human-capital composition term by $1/\bar{n}$. These forces make sex selection more intense in levels under a tight fertility cap.

The marginal effect of a maternal wage increase is different. In the unconstrained model, the fertility-price channel ($\gamma \cdot \partial b^S / \partial p_n$) is a primary positive force driving b up when w_m rises: higher maternal wages make another child more costly, fertility falls, and son preference is concentrated over fewer births. When n is fixed at $\bar{n}$, the time cost $w_m \gamma \bar{n}$ becomes a fixed expenditure for the constrained number of births. Since households cannot reduce n in response to that higher cost, the fertility-price channel is shut off. The remaining forces are the leisure-price channel, the sex-selection price channel (from changes in the education gap), the income effect, and the value channel from daughters' returns. Thus the constrained model can have a higher level of sex selection even though the marginal maternal-wage effect is not necessarily larger than in the unconstrained model.

Second, the relative-returns channel changes as well. When province-level returns r_g rise independently of w_m (as in the preceding subsection), the cost and value channels still oppose each other (Equation A29), but the constrained setting changes their relative force. Because n is fixed, higher daughter investment no longer raises the price of an additional child; the remaining cost-

side effect operates mainly through the sex-selection price p_B . At the same time, investment per child is high, so the return to a daughter $r_g h(e_g^*)$ is large. A marginal increase in r_g therefore produces a larger increase in daughters' economic value, amplifying the value channel and making a negative effect of r_g on sex selection more likely.

Third, the income effect on sex selection is likely positive. In the unconstrained model, non-labor income I has a negligible effect on b because it does not enter any effective price. Under constrained fertility, the income gain that would have raised n cannot do so. If sons are income-elastic at the margin ($\eta_B^W > \eta_n^W$), part of the blocked fertility response may instead appear as higher b .

This constrained-fertility case also clarifies how to read the China evidence. With a low fertility cap, parents cannot respond to son preference by having more children until a son is born. The remaining adjustment margin is the sex composition of the permitted births. This is why evidence on China's one-child policy links tighter fertility restrictions to higher sex ratios (Li et al., 2011; Ebenstein, 2011). The same logic makes income shocks more likely to affect sex selection: when additional income cannot raise fertility, it can instead raise the demand for sons, consistent with Almond et al. (2019). Qian's result points in the opposite direction because the shock is different. Higher tea prices increased returns in a female-intensive agricultural activity, raising girls' relative economic value rather than the maternal time cost of childrearing.¹⁷ Under constrained fertility, that value channel can be especially powerful because each child receives more parental attention and investment. The contrast with Vietnam is therefore not a contradiction: Vietnam has an active fertility margin and a maternal time-cost shock, while the China results come from settings where fertility is constrained or where the shock changes girls' relative economic value more directly.

A.2.3 Policy Implications: Childcare versus Education Prices

The effective prices clarify why the model yields a cleaner policy implication for childcare than for education prices. The maternal childcare parameter γ enters only the fertility price, $p_n = w_m \gamma + p_e e_g$. Reducing γ therefore directly weakens the channel through which higher maternal wages make each child more costly, reduce fertility, and intensify sex selection.

The education price p_e is less direct because it affects two prices at once. It enters the fertility price p_n , so a higher education price raises the cost of an additional child. It also enters the sex-selection price, $p_B = p_b + p_e(e_b - e_g)$, because replacing a daughter with a son changes the education bill when boys and girls receive different investment. When $e_b > e_g$, lowering p_e makes children cheaper but also lowers the education-cost penalty for selecting a son. Optimal investments e_b^* and e_g^* also adjust when p_e changes. The model therefore does not imply that lowering education prices is an equivalent substitute for reducing maternal childcare time. Education costs

¹⁷Qian (2008) studies sex-specific agricultural income shocks in rural China, using tea as a female-intensive crop and orchards as a male-intensive crop. The mechanism is therefore not an education-return shock in the narrow sense, nor does it require that mothers' wages rise one-for-one with tea prices. In the notation here, it is best interpreted as a shock to the relative economic value of girls, holding the maternal childcare time cost conceptually distinct.

help explain why maternal wage gains can amplify the fertility response, but the cleaner policy lever in this framework is the maternal time cost of children.

B. Data Appendix: Family Interrelationship Algorithm

Variables categorizing family interrelationships within the household are crucial in this paper. In particular, we need “pointer” variables identifying each person’s mother, father, and spouse. VHLSS has a variable that classifies each individual’s relationship to the household head. However, this variable is ambiguous and erroneous for the following reasons. First, extended family is common in Vietnam: roughly 34.91% of the VHLSS sample lives in the same household with relatives other than their nuclear family members. Because VHLSS does not identify subfamilies within the extended household, the “relate-to-head” variable will miss the interrelationship of subfamilies other than the head’s. Second, with the exception of VHLSS 2002, all other waves do not identify “children-in-law.” An immediate consequence is missing spousal links and therefore parental links in many cases. Thus, we write an algorithm to improve the precision of parent-child and spousal pairings in VHLSS.

We adapt a similar algorithm that has been used by the Minnesota Population Center ([Sobek and Kennedy, 2009](#)) to generate “pointer” variables for IPUMS–International Census Data. We utilize the four variables: relationship to household head, age, and marital status, in combination with the relative position of household members in the roster listing, to infer relationships. We first establish spousal links (generate variable SPLOC); then we find the mother and father for each individual (generate variables MOMLOC and POPLOC). Regarding parental linkage, we look for the mother before looking for the father. Once we find the mother for a child, we assign her husband as the father of this child. Only when we cannot find a child’s mother do we locate the child’s father and assign the father’s wife as the mother.

Tables [B1](#) and [B2](#) document the rules we apply to these matching tasks. Our algorithm applies the rules sequentially. If the first rule finds a match for a given individual, the second rule no longer applies to that person, and so on. Whenever there are ambiguous multiple potential spouses or multiple potential parents, we drop the entire household from the sample.

Fortunately, VHLSS 2014 and 2016 provide variables that identify the biological fathers and mothers of children under 16. We use this information to validate our algorithm and report the results in Table [B3](#). The algorithm identifies both biological parents correctly for 88.95% of children. Looking at each parent separately, it identifies the correct mother for 93.23% of children and the correct father for 90.99%, allowing the other parent link to be incorrect. The joint match rate is mechanically lower than the single-parent match rates because both parent links must be correct at the same time. These validation results suggest that the algorithm recovers parent-child and spousal links with sufficient accuracy for the empirical analysis.

Table B1: Rules for SPLOC Construction

Rule	Individual's relationship to head	Partner's relationship to head	Age difference	Both Married	Require adjacency	Only applicable to 2002	Notes
Strong couple pairing, couple adjacency preferred							
	Head	Spouse	No	No	No		1
	Parent	Parent	No	Yes	Yes		1
	Grandparent	Grandparent	No	Yes	Yes		1
	Child	Child-in-law	No	Yes	Yes	Yes	1
Weak couple pairing, couple adjacent							
	Grandchild	Grandchild	Yes	Yes	Yes		1, 2
	Other relationship	Other relationship	Yes	Yes	Yes		1, 2
	Sibling	Sibling	Yes	Yes	Yes	Yes	1, 2
	Grandchild	Other relationship	Yes	Yes	Yes		1, 2
	Sibling	Other relationship	Yes	Yes	Yes	Yes	1, 2
	Child	Other relationship	Yes	Yes	Yes		1, 2
Weak couple pairing, special type child-child							
	Child	Child	Yes	Yes	Yes		1, 2, 6
Weak couple pairing, couple not adjacent							
	Child	Child-in-law	No	Yes	Closest proximity	Yes	5
	Child	Other relationship	Yes	Yes	No		2, 3
	Child	Child	Yes	Yes	No		1, 2, 4, 6

Notes:

1. Drop the entire household if there is any person that could be assigned to 2 couples by the adjacency rule.
2. A woman can be no more than 20 years older or 35 years younger than a potential male partner.
3. For non-adjacent couple pairing, among the potential spouses who satisfy the age and marital requirements, select the person who is closest in age and impose that the husband is older than the wife.
4. For child-child non-adjacent couple pairing, drop all households where this rule yields multiple potential spouses.
5. These non-adjacent couples are matched based on having closest proximity to each other.
6. Once the couple is identified, the in-law is distinguished. For child-child couples, assume the first listed spouse is the biological child to the household head, hence the second listed spouse is the child-in-law.

C. Additional Tables and Figures

Table B2: Rules for Child-Parent Relationship Construction (after SPLOC is Generated)

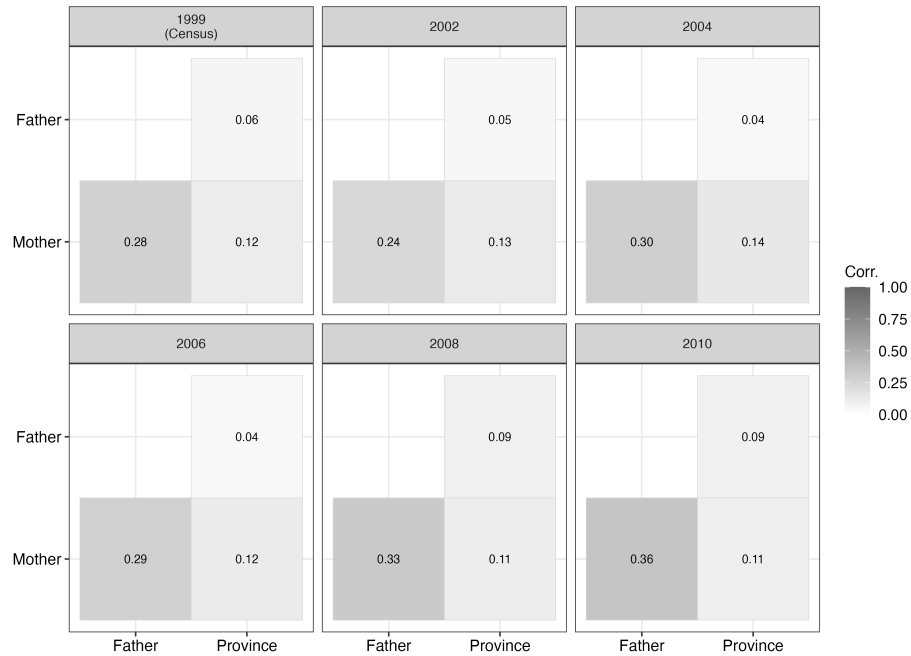
Rule	Child's relationship to head	Parent's relationship to head	Age difference	Proximity requirement applicable	Only applicable to 2002	Notes
Links involving Head, Spouse, and Grandparent (unambiguous)						
	Child	Head, spouse	No	No		
	Child-in-law	Head, spouse	No	No	Yes	4
	Head	Parent	No	No		
	Spouse	Parent	No	No		
	Sibling	Parent	No	No	Yes	
	Parent	Grandparent	No	No		
Links between grandchildren and children						
	Grandchild	Child, child-in-law	15–44	Weak		1
	Grandchild	Other relationship	15–44	Weak		1, 2
Links involving other relatives						
	Other relationship	Grandchild	15–44	Weak		1
	Other relationship	Other relative	15–44	Weak		1
	Other relationship	Sibling	15–44	Weak		1
	Other relationship	Child, child-in-law	15–44	Weak		1, 3
Notes:						
1. Weak proximity requires that the child must be listed after its potential mothers (or potential fathers if it has no potential mothers). Among them, its mother is the one listed closest to the child.						
2. Impose that no person with code "Child" is present in the household. The mother of the grandchildren in these cases tend to be listed as "Others" since there is no category for "Child-in-law."						
3. Impose that no person with code "Grandchild" is present in the household. These cases tend to mix up the numerical code for "Grandchild" (code 6) and that for "Other relationship" (code 7).						
4. The head and spouse are the child-in-law's parents-in-law, not biological parents. This link identifies the household structure (the child-in-law married into the head's family) rather than a biological parent-child relationship. Children-in-law who have their own infants in the household are matched to those infants through the grandchild rules below.						

Table B3: Performance of our algorithm in locating parents

	Freq.	Percent	Cum.
Both mom and dad correct	80021	88.95	88.95
Dad correct, mom incorrect	1831	2.04	90.99
Dad incorrect, mom correct	3853	4.28	95.27
Both mom and dad incorrect	4253	4.73	100.00
Total	89958	100.00	

Notes: This table compares the parent links generated by our algorithm with the biological parent links provided by VHLSS 2014–2016 for children under 16.

Figure C1: Correlations between Treatment Variables by Survey Wave



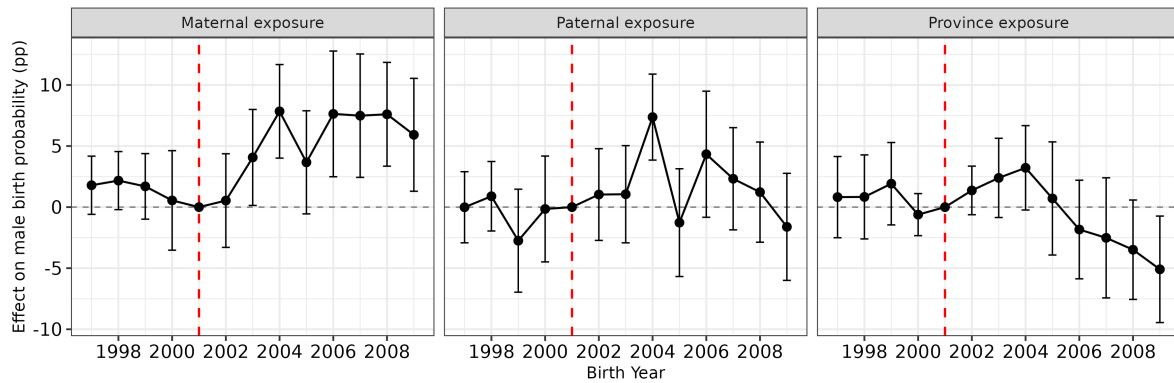
Notes: Pairwise correlations between continuous tariff-cut exposure measures by survey wave. Mother (τ_m) is the mother's industry tariff cut, Father (τ_d) is the father's industry tariff cut, and Province (τ_p) is shift-share provincial exposure. Data are from the 1999 Census baseline and VHLSS 2002–2010.

Table C1: Comparison of Panel Sub-sample and Main Cross-section (2002 Baseline)

	Cross-section (2002 VHLSS)	Panel sub-sample (2002 wave)
Mother's age	28.04 (5.07)	29.05 (5.08)
Mother's education (yrs)	6.61 (3.62)	6.49 (3.60)
Ethnic minority	0.230 (0.421)	0.233 (0.423)
Urban	0.173 (0.378)	0.163 (0.369)
Province exposure	0.09 (0.02)	0.09 (0.02)
<i>N</i>	5,934	879

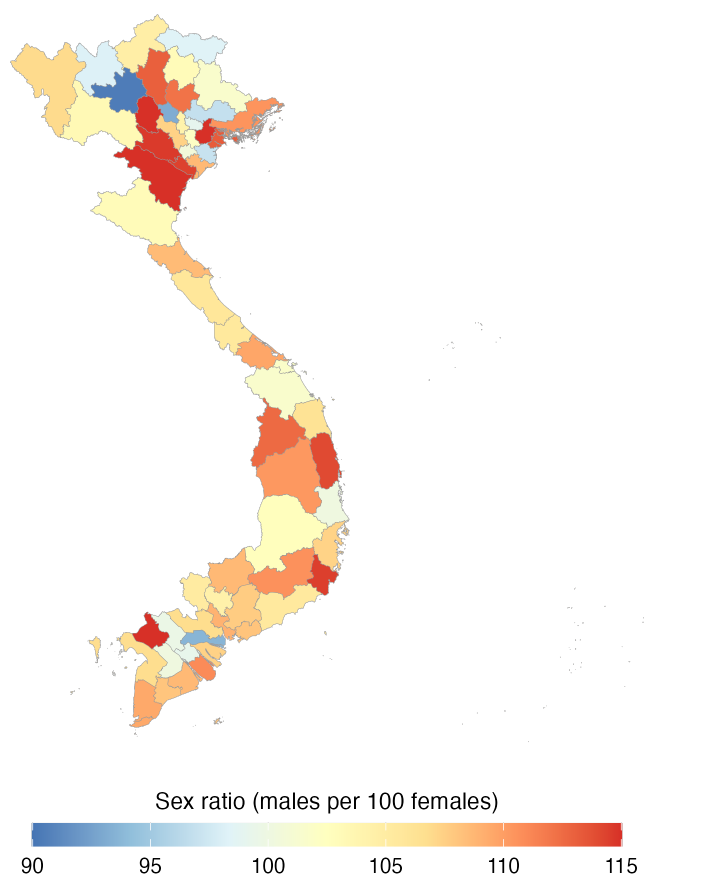
Notes: Standard deviations in parentheses. Cross-section uses 2002 VHLSS mothers of infants aged 0–1. Panel sub-sample uses 2002-wave observations from the rotating-panel sex-selection sample.

Figure C2: Individual Event Study Estimates of BTA Exposure Effects on Male Births



Notes: Separate event study regressions of MaleBirth on year dummies interacted with the indicated continuous exposure for maternal (left), paternal (center), and provincial (right) tariff cut. Parent coefficients are effects of a 0.10 exposure increase and province coefficients are effects of a one-standard-deviation increase. Each regression includes province, birth-year, and parents' ISIC fixed effects and the full control vector from the main specification. Vertical dashed line marks BTA implementation in late 2001. Error bars show 95% confidence intervals with province-clustered standard errors.

Figure C3: Pre-BTA Sex Ratio at Birth by Province



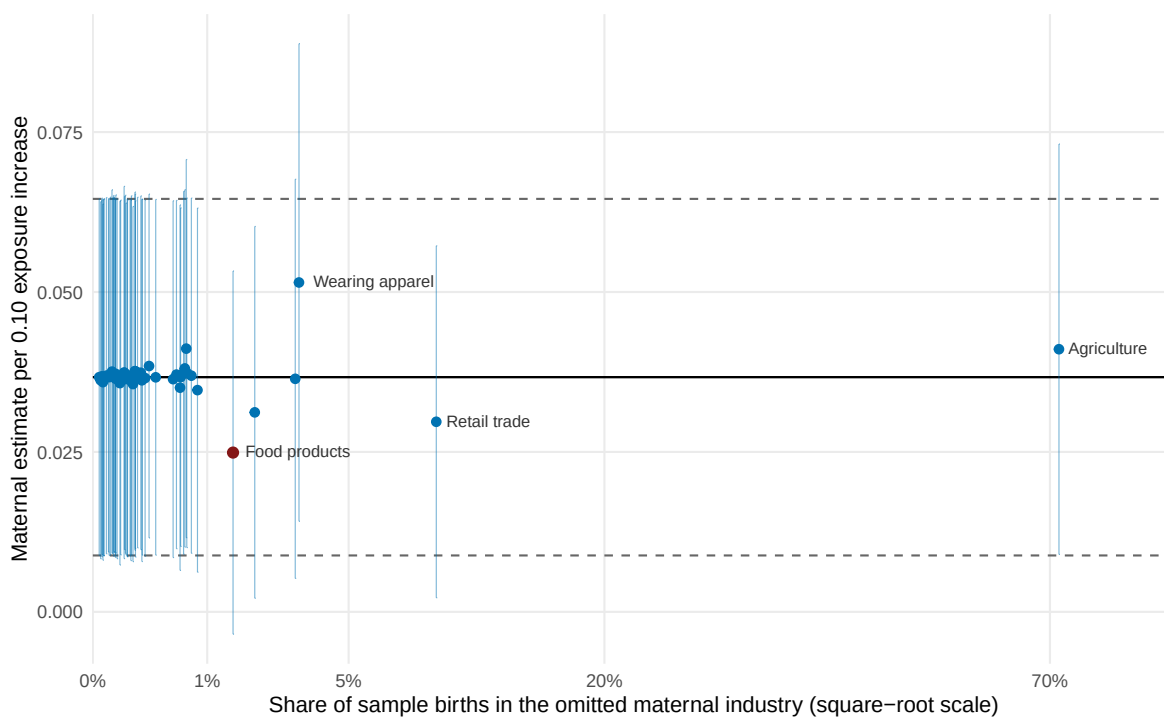
Notes: Sex ratio at birth (males per 100 females) by province from 1997–1998 births in the 1999 Census. Provinces above the median define the high-pre-BTA-SRB subgroup in Figure 5.

Table C2: DHS Benchmark for the Roster-Reconstructed Birth Hazard

	Births within 5 years (1)	Births within 8 years (2)	Births within 10 years (3)	Births within 12 years (4)
VHLSS roster hazard	0.104	0.114	0.122	0.131
DHS birth-history hazard	0.100	0.111	0.120	0.131
Common age-year cells	105	168	210	249
VHLSS women	46,954	52,127	55,024	57,449

Notes: Comparison of the annual birth hazard from the VHLSS 2002 roster reconstruction with the hazard from the pooled 1997 and 2002 Vietnam DHS complete birth histories, for women aged 20–40. Hazards are computed on cells defined by the mother’s age and the calendar year that appear in both sources, and each column restricts births to those occurring within the indicated number of years before the survey. The two hazards agree within 0.4 percentage points on a base of 10 to 13 percent.

Figure C4: Leave-One-Industry-Out Maternal Estimates



Notes: Each point re-estimates the continuous maternal male-birth coefficient, per 0.10 exposure increase, after omitting one maternal industry, plotted against the omitted industry’s share of sample births with vertical bars showing each estimate’s 95 percent confidence interval. The solid line is the full-sample estimate and the dashed lines its 95 percent confidence interval. The red point omits food manufacturing, the most influential industry. All estimates remain positive and inside the full-sample interval.

Table C3: Robustness to ISIC Industry Clustering

	MaleBirth	log(MomWorkHrs)	MomWageEmp	AnyBirth
	(1)	(2)	(3)	(4)
$\tau_m \times \text{Post}$	0.037*** (0.012)	0.033** (0.016)	0.064*** (0.013)	-0.0012 (0.0007)
$\tau_d \times \text{Post}$	0.010 (0.010)	0.016 (0.014)	-0.007 (0.009)	0.0026*** (0.0009)
$\tau_p \times \text{Post}$	0.002 (0.016)	-0.086*** (0.031)	-0.001 (0.024)	0.0038*** (0.0010)
R ²	0.02	0.25	0.60	0.29
Observations	64,170	17,472	17,665	2,011,963
Dep. Var. Mean	0.52	4.68	0.28	0.018
Mom FE				✓
Survey Year FE		✓	✓	
Birth Year FE	✓			✓
Dad ISIC FE	✓	✓	✓	
Mom ISIC FE	✓	✓	✓	
Province FE	✓	✓	✓	
Controls	✓	✓	✓	✓

Notes: Dependent variables, samples, and specifications follow the main Tables 3, 4, and 5. Entries are effects of a 0.10 increase in parent exposure and a one-standard-deviation increase in province exposure. Controls follow the corresponding baseline table: maternal demographics, world-tariff controls, and pre-BTA sector-share-by-year controls where applicable. Standard errors are clustered by 2-digit ISIC industry (59 clusters). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C4: Robustness to Province and ISIC Two-Way Clustering

	MaleBirth	log(MomWorkHrs)	MomWageEmp	AnyBirth
	(1)	(2)	(3)	(4)
$\tau_m \times \text{Post}$	0.037*** (0.012)	0.033 (0.020)	0.064*** (0.015)	-0.0012 (0.0008)
$\tau_d \times \text{Post}$	0.010 (0.011)	0.016 (0.015)	-0.007 (0.008)	0.0026*** (0.0009)
$\tau_p \times \text{Post}$	0.002 (0.010)	-0.086** (0.035)	-0.001 (0.022)	0.0038*** (0.0010)
R ²	0.02	0.25	0.60	0.29
Observations	64,170	17,472	17,665	2,011,963
Dep. Var. Mean	0.52	4.68	0.28	0.018
Mom FE				✓
Survey Year FE		✓	✓	
Birth Year FE	✓			✓
Dad ISIC FE	✓	✓	✓	
Mom ISIC FE	✓	✓	✓	
Province FE	✓	✓	✓	
Controls	✓	✓	✓	✓

Notes: Dependent variables, samples, and specifications follow the main Tables 3, 4, and 5. Entries are effects of a 0.10 increase in parent exposure and a one-standard-deviation increase in province exposure. Controls follow the corresponding baseline table: maternal demographics, world-tariff controls, and pre-BTA sector-share-by-year controls where applicable. Standard errors are two-way clustered by province and 2-digit ISIC. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C5: Reconciliation of Binary and Continuous Male-Birth Estimates

	Binary (1)	Continuous: +0.10 (2)
LPM estimate	0.098***	0.037**
Province-clustered SE	(0.036)	(0.014)
Wild-cluster bootstrap p-value	0.015	0.027
Raw treated-side industries	29	31
Treated-side identifying-mass effective N	6.18	4.61

Notes: Column (1) is the maternal binary high-versus-low contrast (agriculture entered separately) and column (2) the continuous estimate per 0.10 exposure increase, both from the fully controlled male-birth specification. Wild p -values use a restricted province-level Rademacher wild-cluster bootstrap- t with 999 replications. Treated-side effective N is the inverse Herfindahl of identifying exposure-weight mass among treated maternal observations. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C6: Specification Build-Up for the Male-Birth Estimate

	FE only (1)	+ Demographics (2)	+ World tariffs (3)	+ Sector trends (4)
$\tau_m \times \text{Post}$	0.021** (0.009)	0.021** (0.009)	0.037** (0.014)	0.037** (0.014)
$\tau_d \times \text{Post}$	0.005 (0.013)	0.005 (0.013)	0.010 (0.013)	0.010 (0.013)
$\tau_p \times \text{Post}$	-0.002 (0.008)	-0.002 (0.008)	-0.003 (0.008)	0.002 (0.012)
R^2	0.015	0.015	0.015	0.016
Observations	64,170	64,170	64,170	64,170
Birth Year FE	✓	✓	✓	✓
Dad ISIC FE	✓	✓	✓	✓
Mom ISIC FE	✓	✓	✓	✓
Province FE	✓	✓	✓	✓

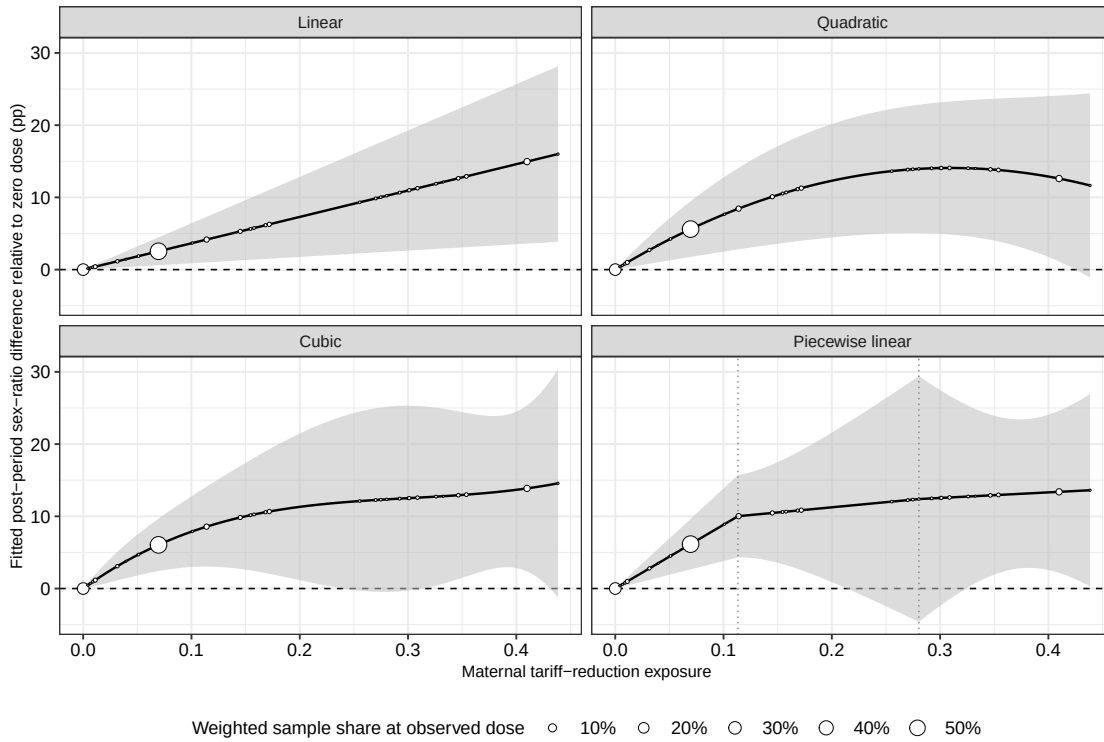
Notes: Dependent variable is MaleBirth. Entries are effects of a 0.10 increase in parent exposure and a one-standard-deviation increase in province exposure. All columns use the fixed full-control estimation sample so coefficient movements are not sample-composition artifacts. Column (1) includes fixed effects only. Columns (2)–(4) add maternal demographics, world-average tariff controls, and pre-BTA sector-share-by-year trends cumulatively. Province-clustered standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C7: Decomposition of Control-Induced Coefficient Movement

	MaleBirth, $\tau_m \times \text{Post}$ (per 0.10)
Coefficient, fixed effects only	0.0209
Coefficient, full controls	0.0368
Gap explained by controls	-0.0159
Contribution of demographics	0.0003
Contribution of world tariffs	-0.0165
Contribution of sector trends	0.0003

Notes: The decomposition follows [Gelbach \(2016\)](#), attributing the gap between the fixed-effects-only and full-control maternal coefficients to each control block through auxiliary regressions of the block's fitted contribution on the treatment interactions. Contributions sum to the gap by construction. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure C5: Dose Sensitivity of the Continuous Male-Birth Estimate



Notes: The figure compares linear, quadratic, cubic, and piecewise-linear dose specifications for the effect of maternal exposure on male birth probability, evaluated relative to a dose of zero. The implied effect at a 0.10 dose is 0.036 in the linear specification and between 0.076 and 0.088 in the flexible specifications. These curves are functional-form sensitivities, not causal dose-response estimates, which would require stronger assumptions than standard parallel trends under heterogeneous effects ([Callaway et al., 2024](#)).

Table C8: Robustness to Including Potentially Endogenous Demographic Controls

	MaleBirth	log(MomWorkHrs)	MomWageEmp	AnyBirth
	(1)	(2)	(3)	(4)
$\tau_m \times \text{Post}$	0.038*** (0.014)	0.035 (0.024)	0.063*** (0.015)	-0.0012** (0.0005)
$\tau_d \times \text{Post}$	0.009 (0.013)	0.015 (0.018)	-0.007 (0.007)	0.0026*** (0.0005)
$\tau_p \times \text{Post}$	0.002 (0.012)	-0.082** (0.039)	-0.002 (0.023)	0.0038*** (0.0011)
R ²	0.02	0.25	0.60	0.29
Observations	63,894	17,472	17,665	2,011,963
Dep. Var. Mean	0.53	4.68	0.28	0.018
Mom FE				✓
Survey Year FE		✓	✓	
Birth Year FE	✓			✓
Dad ISIC FE	✓	✓	✓	
Mom ISIC FE	✓	✓	✓	
Province FE	✓	✓	✓	
Controls	✓	✓	✓	✓

Notes: Dep. vars. are MaleBirth (col. 1), log monthly work hours (col. 2), wage employment (col. 3), and any birth (col. 4). Columns (1)–(3) add demographic controls to the corresponding main-specification control vector: mother's age and parity for MaleBirth, and mother's age for labor outcomes. Column (4) adds mother's age in the birth year; parity is not added to the fertility specification with mother fixed effects. Province-clustered standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C9: Timing and Local-Shock Robustness of the Male-Birth Estimate

	Baseline (1)	Province $\times$ Year FE (2)	Births before 2007 (3)	Post split (4)
$\tau_m \times \text{Post}$	0.037** (0.014)	0.037*** (0.014)	0.038** (0.015)	
$\tau_m \times \text{Post}_{2002-2005}$				0.034** (0.014)
$\tau_m \times \text{Post}_{2006-2009}$				0.064*** (0.021)
$\tau_d \times \text{Post}$	0.010 (0.013)	0.010 (0.014)	0.016 (0.012)	
Observations	64,170	64,170	59,270	64,170
Birth Year FE	✓		✓	✓
Province FE	✓		✓	✓
Province $\times$ Year FE		✓		
Dad ISIC FE	✓	✓	✓	✓
Mom ISIC FE	✓	✓	✓	✓
Controls	✓	✓	✓	✓

Notes: Dependent variable is MaleBirth. Entries are effects of a 0.10 increase in parent exposure. Column (2) replaces the separate province and birth-year fixed effects with province-by-birth-year fixed effects, which absorb the provincial exposure channel. Column (3) restricts to births before 2007. Column (4) splits the post period into 2002–2005 and 2006–2009; the difference between the two maternal coefficients has $p = 0.03$. Province-clustered standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C10: Robustness to Region-by-Year Fixed Effects

	MaleBirth	log(MomWorkHrs)	MomWageEmp	AnyBirth
	(1)	(2)	(3)	(4)
$\tau_m \times \text{Post}$	0.037** (0.014)	0.032 (0.024)	0.061*** (0.015)	-0.0012** (0.0005)
$\tau_d \times \text{Post}$	0.010 (0.013)	0.019 (0.018)	-0.011 (0.008)	0.0026*** (0.0005)
$\tau_p \times \text{Post}$	0.005 (0.011)	-0.075** (0.031)	0.007 (0.015)	0.0038*** (0.0011)
R ²	0.02	0.25	0.61	0.29
Observations	64,170	17,472	17,665	2,011,963
Dep. Var. Mean	0.52	4.68	0.28	0.018
Mom FE				✓
Survey Year FE		✓	✓	
Birth Year FE	✓			✓
Region-Year FE	✓	✓	✓	✓
Dad ISIC FE	✓	✓	✓	
Mom ISIC FE	✓	✓	✓	
Province FE	✓	✓	✓	
Controls	✓	✓	✓	✓

Notes: Continuous exposures (τ_m , τ_d , τ_p) with region-by-year fixed effects added to the baseline specification. Entries are effects of a 0.10 increase in parent exposure and a one-standard-deviation increase in province exposure. MaleBirth and labor outcomes include province and parents' ISIC fixed effects; AnyBirth uses mother fixed effects. Controls follow the corresponding baseline table. Province-clustered standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C11: Robustness to Logit Estimation for Male Birth Outcome

	Continuous exposure		Binary exposure	
	(1)	(2)	(3)	(4)
Maternal exposure $\times$ Post	0.085** (0.035)	0.152** (0.060)	0.286*** (0.092)	0.405*** (0.150)
Paternal exposure $\times$ Post	0.021 (0.052)	0.042 (0.054)	0.041 (0.115)	0.103 (0.123)
Provincial exposure $\times$ Post	-0.007 (0.034)	0.009 (0.051)	0.041 (0.047)	0.067 (0.065)
Implied maternal AME	0.021	0.037	0.070	0.099
Observations	64,162	64,162	64,162	64,162
Birth Year FE	✓	✓	✓	✓
Dad ISIC FE	✓	✓	✓	✓
Mom ISIC FE	✓	✓	✓	✓
Province FE	✓	✓	✓	✓
Controls		✓		✓

Notes: Logit estimates for the male-birth outcome. Columns (1)–(2) use the continuous exposures, with maternal and paternal coefficients per 0.10 exposure increase and the provincial coefficient per one standard deviation. Columns (3)–(4) use the non-agriculture binary contrasts. The implied maternal average marginal effect multiplies the maternal coefficient by the weighted mean of $\hat{p}(1 - \hat{p})$ over the estimation sample. Specifications include birth-year, province, and parents' ISIC fixed effects. Province-clustered standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Maternal exposure denotes the continuous τ_m in columns (1)–(2) and the binary indicator in columns (3)–(4), and likewise for the paternal and provincial rows.

Table C12: Relative>Returns Test for All Outcomes with All Four Exposure Channels

	MaleBirth	log(MomWorkHrs)	MomWageEmp	AnyBirth
	(1)	(2)	(3)	(4)
$\tau_m \times \text{Post}$	0.037** (0.014)	0.034 (0.024)	0.063*** (0.015)	-0.0012** (0.0005)
$\tau_d \times \text{Post}$	0.009 (0.013)	0.021 (0.018)	-0.009 (0.008)	0.0025*** (0.0005)
$\tau_p \times \text{Post}$	0.056 (0.084)	-0.538*** (0.114)	0.169** (0.067)	0.0114** (0.0054)
$\tau_p^w \times \text{Post}$	-0.051 (0.076)	0.414*** (0.094)	-0.157** (0.062)	-0.0070 (0.0043)
R ²	0.02	0.25	0.60	0.29
Observations	64,170	17,472	17,665	2,011,963
Dep. Var. Mean	0.52	4.68	0.28	0.018
Mom FE				✓
Survey Year FE		✓	✓	
Birth Year FE	✓			✓
Dad ISIC FE	✓	✓	✓	
Mom ISIC FE	✓	✓	✓	
Province FE	✓	✓	✓	
Controls	✓	✓	✓	✓

Notes: One controlled specification per outcome. τ_m , τ_d , and τ_p are the continuous exposures from the main analysis, with parent coefficients per 0.10 exposure increase and provincial coefficients per one standard deviation. τ_p^w is the standardized female-weighted provincial exposure ($\sum_j f_j \cdot s_{jp} \cdot \tau_j$, f_j = pre-BTA female employment share), also per one standard deviation. Post = birth year > 2001 (MaleBirth, AnyBirth) or survey year > 2002 (labor). Columns (1)–(3) use the cross-sectional/labor control vector and include province, parents' ISIC, and time fixed effects. Column (4) uses the fertility panel structure from Table 5, with mother and birth-year fixed effects. Province-clustered standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C13: Pre-Treatment Bargaining Power and Sex Selection, 1999 Census

	Mom $\leq$ Dad education	Mom $>$ Dad education
Proportion male	0.5125 (0.0029)	0.5145 (0.0049)
<i>N</i>	30,548	10,619
<i>Regression. Dep. var. = Male birth</i>		
Mom $>$ Dad education		0.0026 (0.0048)
<i>N</i>		41,167
Province FE		Yes

Notes: 1997–1998 births from the 1999 Vietnam Census (IPUMS). “Mom $>$ Dad education” = 1 if mother’s education exceeds father’s. Top panel reports group means (standard errors of means in parentheses). Bottom panel reports OLS with province fixed effects. Province-clustered standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C14: Vietnam Infant Mortality Rate by Sex, 2002–2008

Year	Infant mortality rate			Cumulative % change	
	Male	Female	Gap	Male	Female
2002	23.1	17.4	5.7	—	—
2003	22.3	16.7	5.6	-3.5	-4.0
2004	21.6	16.2	5.4	-6.5	-6.9
2005	20.9	15.8	5.1	-9.5	-9.2
2006	20.4	15.4	5.0	-11.7	-11.5
2007	20.0	15.1	4.9	-13.4	-13.2
2008	19.6	14.8	4.8	-15.2	-14.9

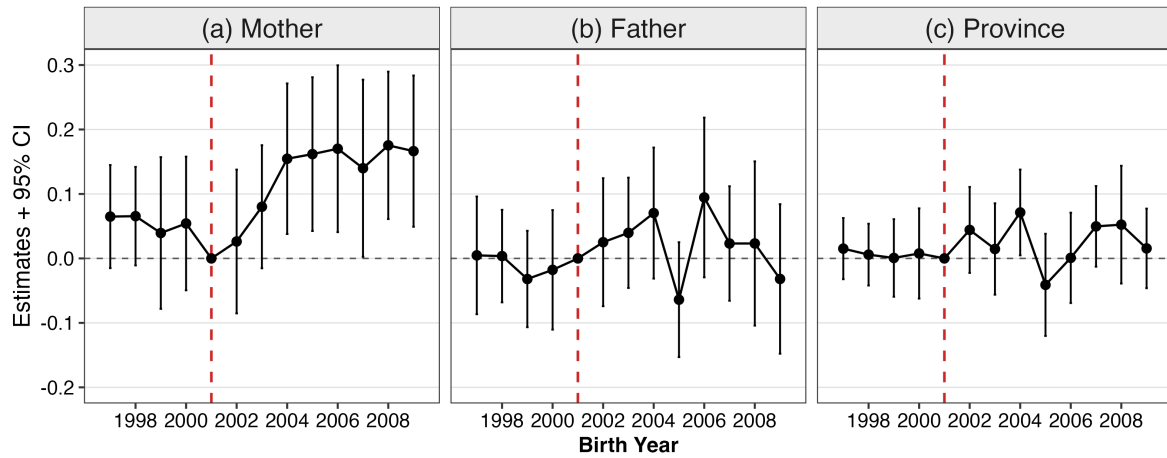
Sources: World Bank, World Development Indicators. Notes: Infant mortality rates per 1,000 live births. “Gap” is male minus female mortality. “Cumulative % change” is relative to the 2002 baseline.

Table C15: Binary-Treatment Contrasts Across Outcomes

	MaleBirth (1)	log(MomWorkHrs) (2)	MomWageEmp (3)	DadWageEmp (4)	AnyBirth (5)
$M_i \times \text{Post}$	0.098*** (0.036)	0.013 (0.057)	0.076*** (0.024)	-0.032 (0.039)	-0.0028** (0.0011)
$F_i \times \text{Post}$	0.025 (0.030)	0.032 (0.042)	0.001 (0.017)	0.047** (0.023)	0.0042*** (0.0011)
$P_p \times \text{Post}$	0.016 (0.016)	0.062 (0.044)	-0.010 (0.023)	-0.024 (0.034)	-0.0006 (0.0017)
R ²	0.02	0.25	0.60	0.53	0.29
Observations	64,170	17,472	17,665	17,665	2,011,963
Dep. Var. Mean	0.52	4.68	0.28	0.44	0.018
Mom FE					✓
Survey Year FE		✓	✓	✓	
Birth Year FE	✓				✓
Dad ISIC FE	✓	✓	✓	✓	
Mom ISIC FE	✓	✓	✓	✓	
Province FE	✓	✓	✓	✓	
Controls	✓	✓	✓	✓	✓

Notes: Dependent variables, samples, controls, and fixed effects follow the continuous main Tables 3, 4, and 5. M_i and F_i equal one if the mother's or father's industry is non-agricultural with a tariff cut strictly above the unweighted industry-level non-agriculture median. P_p is the above-median provincial exposure indicator. Province-clustered standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure C6: Binary-Treatment Event Study of Male Births



Notes: The figure uses the same format as the continuous event study in Figure 4. Parent coefficients compare high and low non-agriculture exposure, and the province indicator uses its median cutoff. The omitted year is 2001, the vertical line marks BTA implementation, and error bars are 95 percent confidence intervals with province-clustered standard errors.